

Dual Sourcing under Internal and External Volatilities

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Abstract

Problem definition: We study a dual sourcing problem in an increasingly volatile world. We consider two types of volatilities. *External volatilities* reflect fluctuating economic conditions via an underlying Markov-modulated state-of-the-world which affects the two suppliers' cost structures, capacity limits, supply mechanisms and demands. *Internal volatilities* affect the actual outputs resulting from random supply processes. Demand distributions are impacted by both types of volatilities. **Methodology/results:** We show how the optimal combined ordering strategy from the two suppliers, along with a salvaging policy, can be efficiently computed, under the conventional assumption of consecutive lead times. We characterize the relatively simple structure of the optimal policies and systematically compare the two types of volatilities. **Managerial implications:** By exploiting the dual sourcing options, we find that the firm can *benefit* from external volatilities; indeed, benefits increase as volatilities increase in specific ways. Numerical studies illustrate these results and reject other reasonable conjectures.

Key words: dual sourcing, random supply mechanisms, external volatilities, internal volatilities

1 Introduction

The global economy is increasingly volatile, forcing companies to continuously revisit their sourcing decisions and associated procurement strategies. Manufacturing labor costs in China have risen sharply over the past four years and are now more than double those in neighboring Vietnam and 35% higher than those in Mexico. The USA, China and Europe have been rapidly imposing, increasing, decreasing or eliminating tariffs in their trade wars. Commodity prices are also increasingly volatile, as are logistical costs. There are, often, significant discrepancies between theoretical capacity levels and actual ones, due to supply disruptions, pandemics, quality problems, rationing schemes, etc.

External volatilities arise because of randomly changing environmental factors, such as fluctuating economic conditions (currency and interest rates, tariffs, labor cost indices, consumer confidence index,

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etc.), seasonal patterns and uncertain market conditions, at different stages of a product life-cycle. These may have a major effect on many primitives of the problem. For such situations, the Markov chain approach for an underlying “state-of-the-world” provides a natural and flexible alternative for modeling factors like the demand process, cost rates, and capacity levels. In such an approach, environmental factors are represented by a “state-of-the-world” modeled to fluctuate according to a Markov chain. For example, the random demand faced in any given period may have a distribution determined by this state-of-the-world. Markov-modulated demand processes also allow for parsimonious ways to represent demands that are correlated across time, contrary to the vast majority of inventory theory which, primarily for analytical tractability, assumes independence across time. Cost parameters, capacity levels and supply mechanisms may, similarly, fluctuate as a function of the state-of-the-world.

Internal volatilities affect, under a given state of the world, the actual demand and also the actual outputs resulting from the random supply processes. Suppliers may only deliver a random fraction of the orders placed. This type of supply risk emerges because of uncertain yields, disruptions such as fires, hurricanes, strikes, sabotages and terrorist attacks, or a random fraction of a production batch clearing quality standards. In other settings, the capacity levels are random, because of uncertain availability of machines, human operators, or critical components or raw materials.

In most supply chain settings, *volatility* is the prime driver of increased operating costs or reduced service levels or sales volumes. In contrast, one of the main managerial insights derived in this paper is that, under dual (or multi-) sourcing, increased external volatilities, far from acting as a hindrance, may, in fact, be exploited to *improve* system performance, by effectively shifting procurement orders between the suppliers, as the state-of-the-world evolves. The potential for such exploitations increases with the available capacity levels.

As stated, we take advantage of the dual (or multi-) sourcing opportunities to drive down the total cost under a more volatile environment. Manufacturing companies and retail chains often have access to two alternative supply sources for component parts, product modules, finished goods or supply materials. One source is typically low cost but has longer lead times, whereas the other provides quicker response but at a higher price. The sources may be capacitated, and their capacity limits may be different. Their supply mechanisms may be random and different. Which of the two sources is more attractive depends heavily on the difference between the variable sourcing costs; however, this difference may fluctuate, heavily and in unpredictable ways, due to the above mentioned sources of volatility. These strategic and tactical dilemmas appear, in particular, when deciding between offshoring and onshoring. The trend towards onshoring had already taken root in the past decade, but the COVID19 pandemic accelerated this trend and the realization that single sourcing results in highly vulnerable supply chains. National industrial policies are reinforcing these trends. For example, the US’ Executive Order on Preserving Essential Medicines (August 4, 2020) mandates both “Maximizing Domestic Production

in Procurement”, as well as (Section 2(ii)) “dividing procurement requirements among two or more manufacturers”. The Executive Order, on America’s Supply Chains (February 2021) calls for more resilient supply chains, facilitating domestic production, built-in redundancies and adequate stockpiles. This Executive Order resulted in the creation of a “Supply Chain Disruptions Task Force”.

The benefits of dual sourcing over single sourcing are often significant, even in a world with stable cost, demand and capacity structures and even under deterministic supplies. See [Xin and Van Mieghem \(2023\)](#) and [Federgruen et al. \(2022\)](#) and the references therein. Not surprisingly, they are even more extensive in settings with fluctuating costs, demand and capacity patterns, and uncertain yields.

[Wagner et al. \(2018\)](#) recently conducted a survey among 1,836 operations managers and executives in 1,000 large manufacturing firms in Europe and the US, covering 17 industries and firms with less than 100 employees (8.6%) as well as those with more than 10,000 employees (40%) and everything in between. The objective was to assess the impact of sourcing flexibility on the firm’s bottom line, through its delivery performance. To this end, they defined sourcing flexibility as a composite measure, composed of high flexibility within supplier contracts, a broad range of possible order sizes from suppliers, frequent changes of volume allocation among existing suppliers and frequent changes of suppliers’ order quantities. The authors concluded that there is a strong correlation between sourcing flexibility and delivery performance and the latter positively related to product financial performance (see Hypotheses 1, 3 and 4).

In this paper we analyze a single-item, dual sourcing, periodic review model, with a finite or infinite planning horizon, in which cost parameters, demand distributions, capacity levels and random supply mechanisms may fluctuate as a function of an underlying state-of-the-world that captures the relevant macroeconomic, seasonal and political factors such as tariffs, the labor cost index, interest and currency rates, etc. This state-of-the-world evolves periodically in accordance with a finite state Markov chain.

Each source, in any given period, has its own cost parameters, lead time, capacity limits for each period’s orders and its own random supply mechanism. Additional costs include inventory carrying and backlogging costs, where we assume that all shortages can be backlogged. Leftover inventory at the end of each period can either be carried over to the next period, incurring an inventory holding cost, or be salvaged, incurring variable salvage revenues. (Salvaging is a frequently used mechanism to reduce high inventory levels, see e.g. [Faulkner \(2023\)](#).) All demand distributions, variable cost and revenue rates, capacity levels and random supply mechanisms may depend on the state-of-the-world.

Contributions. Below, we summarize our main contributions which come in two categories: (I) structural results for optimal strategies, and (II) comparison results for two types of volatilities. Our exact results assume that lead times of the two suppliers differ by one. If the difference is larger than one, no collapse to one-dimensional state spaces is feasible: a minimal state representation requires a vector of inventory measures, the dimension of which is given by the lead time difference Δl . However,

the exact results in this paper suggest effective heuristics when $\Delta l \geq 2$, similar to [Federgruen et al. \(2022\)](#) for systems with deterministic supplies and without Markov modulation, see also Section 7.

(I) *Structural results* (Theorems 1 to 5 and Corollaries 1 and 2), see also Table 1, where our contributions are contrasted with those in the most general existing literature on multi-sourcing.

- (i) Parallel ordering (general supply mechanisms): *Optimality of a common threshold policy*. We first analyze the model assuming that in each period orders need to be placed in parallel. This is the most prevalent assumption in the literature that deals with random supplies; see, however, (ii) and (iii) below. We give a simple sufficient and necessary condition for the optimality of a *common* threshold policy: if the period's starting inventory position is below [above] a first [second] threshold, it is optimal to place an order [salvage] (Theorem 1). When salvaging is not an option, the condition is always satisfied (Corollary 1). Theorem 1 characterizes the structure of an optimal strategy under arbitrary parameter and distributional assumption.
- (ii) Sequential ordering (general supply mechanisms): *Optimality of individual threshold policies for the suppliers*. There are many settings where one can observe the actual realization of one of the orders, before determining the second order (see Section 4.2). Exploiting this information can, of course, reduce the expected costs. Moreover, even more specific structural results can be obtained in this case. When orders may be placed sequentially, a *first*-stage individual threshold policy determines how any expected order and any salvage quantity are to be chosen: when the starting inventory position is below [above] a first [second] threshold value, an expedited order [salvage batch] needs to be placed [initiated]. A similar *second*-stage threshold policy determines any order with the regular supplier, based on the *adjusted* inventory position = inventory position plus net addition = inventory position + expedited order – salvage batch (Theorem 2).
- (iii) Sequential ordering (random capacity model): *Optimality of individual modified base-stock policies*. When the supply mechanism follows the random capacity model (type (4a)), i.e., the theoretical capacity is capped by a random variable, the above threshold policies become modified based-stock policies: one brings the relevant inventory position as close as possible to the associated base-stock level, given the capacity levels (Theorem 3 and Corollary 2, the latter dealing with the case where no salvaging option exists).
- (iv) Deterministic supply mechanisms: While the literature on dual sourcing has almost exclusively focused on this setting, it can actually be considered as a special case of the previous case. (Under deterministic supplies, sequential ordering can be assumed w.l.o.g.) Beyond the special structure identified under (iii), we are able to provide a precise characterization of how the total order is allocated among the two suppliers, or among the regular supplier and salvaging, and how the various model primitives impact on this allocation. (Under deterministic supplies, expedited ordering and salvaging are mutually exclusive.) See Appendix B.

- (v) Monotonicity with respect to the state-of-the-world: Assuming only the cost rates [capacity levels] are state-dependent, and that they are uniformly ranked in ascending [descending] order of the state-of-the-world, we show that all value functions are increasing in the state-of-the-world (Theorems 4 and 5 respectively).

In a setting where the suppliers are differentiated by the *lead times* (as well as cost rates, capacity levels and supply mechanisms), a *different* optimality structure is valid as described in Theorem 1. First and foremost, the optimality of any kind of threshold policy depends on a *necessary and sufficient* condition (see Theorem 1). The fundamental difference is that under *identical* lead times, the partial derivatives of each period's minimand in the dynamic program (with respect to the targeted order sizes) are identical but for the per-unit cost rates of the suppliers. Under lead time differences, they differ structurally: the partial derivative with respect to the expedited order depends on the expected holding and backlogging cost functions $\{L_n^i(\cdot)\}$, defined later, where the partial derivative with respect to the regular supplier shows no such dependence. Thirdly, the proof techniques in Feng and Shanthikumar (2018) are very different from those applied here.

Most of the literature on *multi-sourcing* assumes *identical* lead times. Feng and Shanthikumar (2018) set out to prove the optimality of individual threshold policies. However, our policy structure and analysis differ from theirs. First, in our setting where the suppliers are differentiated by the *lead times* (as well as cost rates, capacity levels and supply mechanisms) and with additional salvage opportunities, a *different* optimality structure is valid as described in Theorem 1. In particular, the optimality of any kind of threshold policy depends on a *necessary and sufficient* condition (see Theorem 1). Second, the analyses under identical vs non-identical lead times differ structurally. Under *identical* lead times, the partial derivatives of each period's minimand in the dynamic program (with respect to the targeted order sizes for different suppliers) are identical but for the per-unit cost rates of the suppliers. Under *non-identical* lead times, the partial derivative with respect to the expedited order depends on the expected holding and backlogging cost functions $\{L_n^i(\cdot)\}$, defined later, where the partial derivative with respect to the regular supplier shows no such dependence. Third, the analysis in Feng and Shanthikumar (2018) contains errors, detailed in Online Appendix OA.1. (In addition, it requires additional technical conditions that are hard to verify). A valid analysis was obtained in Federgruen et al. (2023).

There exists a substantial literature on dual sourcing models with supplier-dependent lead times, but that literature, surveyed in Section 2, by and large, assumes *deterministic* supplies. (Exceptions are Tan et al. (2016) addressing supply mechanisms of type (4a) under only parallel ordering, see Remark 6; Chen et al. (2018), Chen and Gao (2019) and Chen (2017) address random supply mechanisms of type (4a) only, which do not result in tractable algorithms to identify the optimal policies, let alone policies of simple structure.)

- (II) *Two types of volatilities* (Theorem 6 and Propositions 1 to 4). In our model, we encounter two

types of volatilities which we show have different effects on the costs of the system:

- *Internal* volatilities: those embedded in the demand distributions and supply mechanisms.
- (vi) Impact of internal volatilities: These volatilities typically cause costs to *increase*. As an example, we show that expected costs increase when supply mechanisms are replaced by ones that dominate in the convex order (Theorem 6).
- *External* volatilities: those resulting from the fluctuating state-of-the-world.
- (vii) Dependency of model parameters on the state-of-the-world: We compare two systems that are identical except for the way the cost rates of either the expedited or the regular supplier depend on the state-of-the-world, focusing on the former (the results for the latter are analogous). Proposition 1 establishes conditions under which a system with a state-of-the-world-*independent* cost rate for the expedited supplier, is nevertheless more expensive than one in which it fluctuates with the state-of-the-world. Proposition 2 generalizes this by comparing two systems with the expedited supplier’s cost rates in both systems increasing in the state-of-the-world. We establish a partial order among the vectors of cost rates, such that the more volatile cost structure in fact leads to lower expected costs, because of our ability to reallocate orders between the suppliers.
- (viii) The Markov chain governing the state-of-the-world: We compare two systems in which the model parameters are identical except for the Markov chains determining the evolution of the state-of-the-world. We give sufficient conditions on the Markov chains to ensure that all value functions are pointwise larger in one of the two systems. This is obtained both when all cost rates [capacity levels] are increasing [decreasing] in the state-of-the-world, with all other model parameters independent of the state-of-the-world: Propositions 3 and 4, respectively.

Thus, the two types of volatility considered in our paper have different impacts on system-wide costs and inventory investments: those intrinsic to the demand distributions or random supply mechanisms (*internal volatilities*), typically result in an *increase* of the system’s expected costs and inventories, the larger this type of volatility is. In contrast, increased volatility in the environmental factors, i.e., the state-of-the-world (*external volatilities*), can often be exploited to *reduce* expected costs and inventories. This is explained by the fact that the state-of-the-world can be observed before placing orders, while internal volatilities cannot be. A dual sourcing strategy allows for adequate shifts in relative orders between the two sources, an option that does not exist in a single sourcing setting.

The remainder of the paper is organized as follows: Section 2 provides a literature review. In Section 3 we introduce our general model and notation. Section 4 develops our structural results. Section 5 establishes sufficient conditions for the value functions to be monotone in the state-of-the-world. Our analysis of the impact of the two types of volatilities are covered in Section 6. Section 7 concludes the paper with some final observations and generalizations. All proofs can be found in Appendix A.

Table 1: Differences in models from related papers

	Lead time difference	Parallel / Sequential	Salvage option	Random supply	Markov modulation	Ordering policy
Theorem 1	1	Parallel	✓	general	✓	nec. & suff. cond. common threshold
Corollary 1	1	Parallel	×	general	✓	common threshold
Theorem 2	1	Sequential	✓	general	✓	individual threshold policies
Theorem 3	1	Sequential	×	random capacity	✓	modified base-stock policies
Corollary 2	1	Sequential	✓	random capacity	✓	double modified base-stock
Feng and Shanthikumar (2018)	0	Parallel	×	general	×	individual threshold policies
Federgruen et al. (2023)	0	Parallel	×	general	×	individual threshold policies
Tan et al. (2016)	1	Parallel	×	random capacity	×	individual threshold policies
Gong et al. (2014)	1	Parallel	×	proportional yield with Bernoulli shocks	✓	threshold policies
Chen (2017), Chen et al. (2018), Chen and Gao (2019)	general	Parallel	×	random capacity	×	monotonicity properties

2 Literature Review

Fukuda (1964), following three earlier papers, showed that under linear ordering costs and in the absence of capacity limits or salvage opportunities, the optimal policy is a dual base-stock policy: the order with the expedited supplier is governed by a base-stock policy acting on the starting (full) inventory position, while that with the regular supplier is determined by a second base-stock policy acting on the adjusted inventory position that includes the order placed with the expedited supplier. Whittmore and Saunders (1977) showed that under *non-consecutive* lead times, the optimal policy depends on a *vector* of inventory measures, the dimension of which is given by the lead time *difference*. As a consequence, the subsequent literature focused, largely, on heuristic policies in Fukuda’s base model, but with general lead times. See Sun and Van Mieghem (2018) and Federgruen et al. (2022) for recent reviews of this literature. Two most recent contributions to this literature stream are Tang et al. (2024) and Chen and Shi (2021), who develop online algorithms for the stationary base model when the demand distribution is unknown and needs to be learned. The two papers assume that the replenishment policy is to be chosen from two different heuristic policy classes. As the lead time difference grows large, Xin and Goldberg (2018) prove that a simple tailored base-surge policy, an early heuristic, is asymptotically optimal. Under stochastic lead times, Song et al. (2017) and Song et al. (2022) consider dual-sourcing and multisourcing problems, and obtain optimal or near-optimal heuristic policies. Wang et al. (2017)

considers dual sourcing under the complication of total minimum order commitments.

As mentioned, an independent literature stream addresses dual and multi-sourcing settings where suppliers experience *randomness* in their supply processes, but have *zero* (or *identical*) lead times. Within this stream, several papers assume only a supplier-dependent random fraction of each order materializes due to supply disruptions or quality control filtering, see [Federgruen and Yang \(2011\)](#) and the references therein. This stochastic proportionality yield model was generalized by [Dada et al. \(2007\)](#) to cover a broad spectrum of concave relationships between input and (stochastic) output. [Huh and Nagarajan \(2010\)](#) analyzes a class of linear inflation rules under random yield in single-sourcing. [Dong et al. \(2022\)](#) studies the effect of correlated random yields among multiple sources, and considers the procurement strategies in terms of supply diversification. Another set of papers consider settings where a supplier’s *aggregate* capacity is random, see [Chen and Gao \(2019\)](#) and the references therein. [Feng and Shanthikumar \(2018\)](#) synthesized these various literature streams, developing a general random supply framework, but continue to assume all suppliers have identical lead times.

[Chen et al. \(2018\)](#) address a dual sourcing model with random aggregate capacities (and no salvage opportunities) and supplier dependent lead times; see also [Chen and Gao \(2019\)](#) and [Chen \(2017\)](#). The optimal policy acts on a vector of inventory measures, as explained above. Certain monotonicity properties can be derived but they do not result in tractable algorithms to identify the optimal policy, let alone policies of simple structures. [Chen et al. \(2022\)](#) study a joint production and inventory model with two kinds of input raw materials under availability uncertainties. [Yazlali and Erhun \(2007\)](#) considers a multisourcing setting with supplier-dependent lead times, but deterministic supply mechanisms; the authors show that a modified base-stock policy for each supplier is optimal; this result is generalized in Corollary 2 to allow for general random capacities. The case of parallel ordering was addressed by [Tan et al. \(2016\)](#). These authors show that in this case, modified base-stock policies may fail to be optimal, consistent with our structural results in (i) above; see, however, Remark 6.

The paradigm of Markov-modulated inventory systems was initiated in seminal periodic review papers by [Karlin and Fabens \(1960\)](#) and [Iglehart and Karlin \(1962\)](#), but, almost invariably for single supplier systems, and under deterministic supplies. See also [Zipkin \(2000\)](#).

To our knowledge, [Gong et al. \(2014\)](#), building on [Chao et al. \(2008\)](#), is the only paper addressing a Markov-modulated dual sourcing problem, indeed one with random supplies. Each supplier may be in an “on” or “off” state; the two supplier states evolve according to two independent two-state Markov chains. The suppliers’ states in a given period become known only after orders are placed. The model may be viewed as a special case of ours, without capacity limits or salvage opportunities, lead times $l_e = 0$ and $l_r = 1$ and random but independent supply processes which follow the proportional yield model with Bernoulli yield distributions. The authors allow for price-dependent demand distributions with each period’s price chosen by the firm. In our model, we omit this additional inventory and profit

lever, but it could be added with minor adjustments.

3 Model and Notation

We index the periods backward from 1 to N . The sequence of events in period n is as follows: at the beginning, an order may arrive from the expedited or the regular supplier, or both. Such deliveries are added to the inventory level. The Markovian state-of-the-world is then revealed. The firm now decides on new order sizes to be placed with each of the two suppliers as well as any salvage quantity if it wants to *reduce* its inventory position. The *actual* order supply is a random function of the order size, realized within the period. (As mentioned, we systematically distinguish between parallel and sequential ordering by the two suppliers.) The inventory adjustments are realized after their corresponding lead times. After the placement of any new orders or the initiation of any salvage batch, stochastic demand is then realized and satisfied with on-hand inventory. At the end of the period, any unsatisfied demand is fully backlogged while leftover inventory is carried over to the next period.

Random supply mechanisms arise, sometimes, because the supplier's theoretical capacity level is subject to random shocks, jeopardizing in particular larger orders that come close to requiring all of the supplier's capacity. In other settings, all orders may be curtailed because only a random yield fraction comes through. This applies, for example, to agricultural processes, where only a random fraction of the planted acreage is harvested due to random weather, rain and irrigation conditions. Random yield factors also apply due to quality control procedures clearing only a random fraction of the production batch. Finally, such random yield factors arise due to complete or partial supply disruptions, themselves the result of strikes, fires, pandemics, etc. We provide a general treatment for all such random supply mechanisms, following [Feng and Shanthikumar \(2018\)](#), making minimum requirements for these mechanisms.

For period $n = N, \dots, 1$, let

l_e, l_r, l_v = the lead time for ordering from the expedited supplier, regular supplier, salvaging, respectively, where $l_r = l_e + 1$ and $l_v = l_e$,¹

i_n = the state-of-the-world in period n , $i_n \in \mathcal{I} := \{1, \dots, I\}$;

$\{i_n\}$ = an I -state Markov chain with $I \times I$ transition matrix $P = \{p_{ij}\}$;

$c_{n,e}^i, c_{n,r}^i$ = unit price charged by the expedited and regular suppliers for orders placed in period n , respectively, when $i_n = i$;

$C_{n,e}^i, C_{n,r}^i$ = capacity limit for any expedited and regular order placed in period n , respectively, when $i_n = i$;

$c_{n,v}^i, C_{n,v}^i$ = unit revenue and capacity limit for salvage batches initiated in period n , respectively, when $i_n = i$;

D_n^i = demand in period n when $i_n = i, D_n^i \geq 0$ with known continuous distribution;
 $Y_{n,e}^i(q), Y_{n,r}^i(q)$ = random supply from the expedited and regular supplier in period n , respectively,
 when placing an order of size q and $i_n = i$;
 α = discount factor, $\alpha \in [0, 1]$.

For any function $h : \mathbb{R} \rightarrow \mathbb{R}$, let $h'(\cdot)$ [$h'^+(\cdot), h'^-(\cdot)$] denote its derivative [right-hand derivative, left-hand derivative]. Similarly, for $h : \mathbb{R}^m \rightarrow \mathbb{R}$ with $m \geq 2$, $\partial^+ h$ [$\partial^- h$] denotes the right-hand [left-hand] partial derivative.

We impose the following restrictions on the cost parameters:

Assumption 1. *In any period, the unit order costs and the unit salvage revenue satisfy the restrictions:*

(i) $c_{n,e}^i \geq c_{n,v}^i$ for any $i \in \mathcal{I}$, and (ii) $\min_{i \in \mathcal{I}} c_{n,r}^i \geq \alpha \max_{j \in \mathcal{I}} c_{n-1,v}^j$.

Assumption 1 (i) precludes arbitrage opportunities when goods are procured from the expedited supplier and sold at a premium via the salvage channel. Similarly, Assumption 1 (ii) precludes the possibility of buying units from the regular supplier and initiating the sell-off of these units, at a profit, one period later. For period $n = N, \dots, 1$, let

x_n = the inventory position at the beginning of period n

= the beginning inventory level (i.e., on-hand inventory minus backlogs), plus all outstanding orders from both suppliers and salvage batches, *exclusive* of this period's new orders;

$q_{n,e}$ = the order with the expedited supplier in period n , ($q_{n,e} \geq 0$);

$q_{n,r}$ = the order with the regular supplier in period n , ($q_{n,r} \geq 0$);

$q_{n,v}$ = the *negative* inventory adjustment due to a salvage batch initiated in period n , ($q_{n,v} \leq 0$).

Assume the inventory and backlogging related costs depend on the end-of-period inventory level only. Instead of charging the actual inventory costs that arise at the end of period n ($n = 1, 2, \dots$) to *that* period, one obtains an equivalent representation of the controllable part of the total expected discounted cost by charging its expected value to period $n + l_e$. In particular, we employ the following two stochastic identities for the system dynamics:

$$x_{n-1} = x_n + q_{n,v} + Y_{n,e}^{i_n}(q_{n,e}) + Y_{n,r}^{i_n}(q_{n,r}) - D_n^{i_n},$$

while the *inventory level* IL_{n-l_e} at the *end* of period $n - l_e$ is given by

$$IL_{n-l_e} := x_n + q_{n,v} + Y_{n,e}^{i_n}(q_{n,e}) - D_{n,n-l_e}^{i_n}, \quad (1)$$

where $D_{n,m}^{i_n}$ is the aggregate demand in the time interval $[n, m]$ with $m \leq n$, when the initial state-of-

¹The case $l_v = l_r$ may be handled analogously.

the-world in period n is i_n , i.e., $D_{n,m}^{i_n} = D_n^{i_n} + D_{n-1}^{i_{n-1}} + \dots + D_m^{i_m}$. For all $n = N, \dots, 1$ and all $i \in \mathcal{I}$, let

$L_n^i(x_n + q_{n,v} + Y_{n,e}^i(q_{n,e}))$ = the *expected value* of all inventory and backlogging related costs at the end of period $n - l_e$ discounted back to period n , when $i_n = i$.

In terms of the expected holding and backlogging cost functions $L_n^i(\cdot)$, we introduce a standard assumption:

- Assumption 2.** (a) The functions $L_n^i(\cdot)$ are strictly convex, with derivatives $L_n^{i'}(\cdot)$ and inverse derivatives $(L_n^{i'})^{-1}(\cdot)$. $L_n^i(x) = O(|x|^p)$ and $L_n^{i'}(x) = O(|x|^{p-1})$ for some integer $p \geq 1$, $n = N, \dots, 1$, $i \in \mathcal{I}$.
- (b) All demand distributions are continuous with $\mathbb{E}[|D_n^i|^p] < \infty$ for some integer $p \geq 1$, $n = N, \dots, 1$, $i \in \mathcal{I}$.
- (c) $\lim_{x \downarrow -\infty} L_n^{i'}(x) < -c_{n,e}^i - \bar{c}_{n-1,r}/\alpha$ and $\lim_{x \uparrow \infty} L_n^{i'}(x) > (-c_{n,v}^i)^+$, where $\bar{c}_{n,r} := \max_{j \in \mathcal{I}} \{c_{n,r}^j\}$.

Remark 1. Let $\tilde{L}_n^i(x)$ denote the holding and backlogging cost as a function of the prevailing inventory level x in period n when $i_n = i$. Note that $L_n^i(x) = \mathbb{E}\tilde{L}_n^i(x - D_{n,n-l_e}^i)$. Note that the random variables $\{D_{n,n-l_e}^i\}$ have continuous distributions because all $\{D_n^i\}$ do. Let $f(x)$ denote its pdf. If $\tilde{L}_n^i(\cdot)$ is itself strictly convex and differentiable, this property is maintained for $L_n^i(\cdot)$. Even if it is not strictly convex or differentiable everywhere, as in the standard specification $\tilde{L}_n^i(x) = h_n^i x^+ + p_n^i x^-$, $L_n^i(\cdot)$ becomes strictly convex and differentiable everywhere, and its second derivative $L_n^{i''}(x) = (h_n^i + p_n^i)f(x) > 0$, assuming the distributions $\{D_n^i\}$ have unbounded support. (If they all have bounded support, they can be approximated arbitrarily closely by ones that have unbounded support.)

Remark 2. While we allow the $L_n^i(\cdot)$ functions to be super-linear, Assumption 2 (a) ensures that they remain polynomially bounded.

Remark 3. To motivate Assumption 2 (c), when $L_n^i(\cdot)$ are super-linear, i.e., $p > 1$, $\lim_{x \downarrow -\infty} L_n^{i'}(x) = -\infty$ and $\lim_{x \uparrow \infty} L_n^{i'}(x) = \infty$ and Assumption 2 (c) is clearly satisfied. When $L_n^i(\cdot)$ are linear, i.e., $p = 1$, as in the standard single-item model, $\lim_{x \downarrow -\infty} L_n^{i'}(x) = -p_n^i \ll -c_{n,e}^i - \bar{c}_{n-1,r}/\alpha$ in all practical instances, while $\lim_{x \uparrow \infty} L_n^{i'}(x) = h_n^i > (-c_{n,v}^i)^+$ when $c_{n,v}^i \geq 0$, i.e., there is positive salvage value.

The remaining case has $c_{n,v}^i < 0$, in which case we want to ensure that salvage, i.e., disposal costs are not so excessive as to make the salvage option prohibitively expensive. In this case, $\lim_{x \uparrow \infty} L_n^{i'}(x) = h_n^i > (-c_{n,v}^i)^+$ may be violated. The purpose of the second limit in part (c) is to ensure that for large enough inventory positions, it is beneficial to use the salvage option when it is available; indeed, when $c_{n,v}^i < 0$ is sufficiently negative, it may not be optimal to use this option. Under stationary salvage cost rates, this happens when $(-c_{n,v}^i) \geq \sum_{r=0}^{n-1} \alpha^r h_{n-r}^i$. Only minor adjustments to the analysis are needed to address this relaxed assumption.

As mentioned in Section 2, various random supply structures have typically been analyzed in isolation. Feng and Shanthikumar (2018) recently showed that a unified treatment can be provided for general random supply functions that map an order of size q into a random variable $Y(q)$. We assume that $Y(q)$ can be represented with a specific functional form involving q and a random variable Z , i.e., $Y(q) := \phi(q, Z)$, with $\phi(\cdot, \cdot)$ strictly increasing and differentiable in q almost everywhere, and the random variable Z with a general distribution with bounded support. Stochastic increasingness is hence implied, as in (3). Also $Y(0) = 0$.

Often, the analysis is greatly facilitated when representing the random supply as a function of the *expected* or targeted supply $\mu \equiv \psi(q) := \mathbb{E}Y(q) = \mathbb{E}_Z \phi(q, Z)$, rather than the order size q . Since $\phi(\cdot, \cdot)$ is strictly increasing in its first argument, so is $\mu = \psi(q) = \mathbb{E}_Z \phi(q, Z)$ strictly increasing in q . This implies the existence of a (strictly increasing) inverse function $q(\mu) = \psi^{-1}(\mu)$, such that $\mu = \mathbb{E} \phi(\psi^{-1}(\mu), Z)$. By the implicit function theorem and the continuous differentiability of ϕ , $q(\mu)$ is differentiable with

$$q'(\mu) = \frac{1}{\mathbb{E}_Z \frac{\partial \phi(q(\mu), Z)}{\partial q}} > 0. \quad (2)$$

Thus, let $\hat{Y}(\mu) := Y(q(\mu)) = \phi(q(\mu), Z)$ denote the random supply as a function of its mean (or targeted) supply μ . Note, by (2) and since the $\phi(q, Z)$ functions are increasing in q :

$$\{\hat{Y}(\mu)\} \text{ is a stochastically increasing family of random variables.} \quad (3)$$

Feng and Shanthikumar (2018) showed that dynamic procurement problems can often be formulated as *convex* dynamic programs, i.e., dynamic programs with convex value functions, if *mean* supplies are used as the action variables and provided the random supply functions $\hat{Y}(\cdot)$ satisfy the so-called *stochastic linearity in midpoint* (SL(mp)) property. (This turns out to be the case in our model, see Lemma 1.)

Definition 1 (Feng and Shanthikumar (2018)). *A family of random variables $\{\hat{Y}(\mu), \mu \in \mathcal{U}\}$ for some convex set $\mathcal{U}$ is stochastically linear in midpoint, written as $\{\hat{Y}(\mu), \mu \in \mathcal{U}\} \in \text{SL}(\text{mp})$, if for any $\mu_1, \mu_2 \in \mathcal{U}$, there exists $\tilde{Y}(\mu_1)$ and $\tilde{Y}(\mu_2)$ defined on a common probability space such that*

- (i) $\tilde{Y}(\mu_i)$ has the same distribution as $\hat{Y}(\mu_i)$, $i = 1, 2$, and
- (ii) $\frac{\tilde{Y}(\mu_1) + \tilde{Y}(\mu_2)}{2} \leq_{\text{concave}} \hat{Y}(\frac{\mu_1 + \mu_2}{2})$.

A pair of random variables (X, Y) is ranked by the concave order $\leq_{\text{concave}}$, if $\mathbb{E}f(X) \leq \mathbb{E}f(Y)$ for any concave function f . It is easily checked that the SL(mp) property is satisfied for any (deterministic) linear function and therefore generalizes deterministic linearity.

Feng and Shanthikumar (2018) exhibited several important random supply structures under which the SL(mp) property is satisfied, see Theorem 5 in Feng and Shanthikumar (2018):

$$(a) \ Y(q) = \min\{q, Z\} \quad (\text{random capacity model}) \quad (4a)$$

$$(b) Y(q) = qZ \quad (\text{stochastic proportional yield model}) \quad (4b)$$

$$(c) Y(q) = qZ/(q + \alpha Z^k) \text{ with } \alpha > 0 \text{ and } k < 1 \quad (\text{see Dada et al. (2007)}) \quad (4c)$$

Feng and Shanthikumar (2018) also provide more general sufficient conditions for the function $\phi(q, Z)$ under which the property $\{\hat{Y}(\mu), \mu \in \mathcal{U}\} \in \text{SL}(\text{mp})$ is guaranteed, in particular for any $q_1 < q < q_2$,

$$\text{the sign of } \phi(q, Z) - \frac{\phi(q_1, Z) + \phi(q_2, Z)}{2} \text{ can change at most once (positive to negative) in } Z. \quad (5)$$

Condition (5), in turn, is guaranteed for three times differentiable functions $\phi(\cdot, \cdot)$, where $\phi(q, Z)$ is concave in q , supermodular in (q, Z) , and $\frac{\partial^3 \phi(q, Z)}{\partial Z \partial q^2} \geq 0$. (See also Chapter 8 of Shaked and Shanthikumar (2007).)

Denote by $Y_{n,e}^i(q_e)$ and $Y_{n,r}^i(q_r)$, and equivalently $\hat{Y}_{n,e}^i(\mu_e)$, $\hat{Y}_{n,r}^i(\mu_r)$, the random supply functions in period n under state-of-the-world i , for the expedited and the regular supplier, respectively. We thus assume:

Assumption 3. For all n, i , the functions $\{\hat{Y}_{n,e}^i(\mu_e), \mu_e \in \mathcal{U}_{n,e}^i\}$ and $\{\hat{Y}_{n,r}^i(\mu_r), \mu_r \in \mathcal{U}_{n,r}^i\}$ are $\text{SL}(\text{mp})$ and independent, where $\mathcal{U}_{n,e}^i = [0, \mathbb{E}\phi_{n,e}^i(C_{n,e}^i, Z_{n,e}^i)]$ and $\mathcal{U}_{n,r}^i = [0, \mathbb{E}\phi_{n,r}^i(C_{n,r}^i, Z_{n,r}^i)]$, with the $\phi(\cdot, \cdot)$ functions strictly increasing and differentiable in q almost everywhere, and the random variables Z with a general distribution with bounded support. Also $\phi(0, Z) = 0$ almost surely.

We first assume that orders to the two suppliers need to be placed *simultaneously* (*parallel ordering*). Thereafter, we address the case where the realization of the random supply by the expedited supplier is revealed *before* the order with the regular supplier needs to be placed (*sequential ordering*).

Define the value function

$f_n^i(x)$ = the optimal discounted expected total cost in the last n periods of the planning horizon, when period n is started with an inventory position x and $i_n = i$.

In terms of the terminal value function $f_0^i(\cdot)$, we merely assume the following.

Assumption 4. $f_0^i(\cdot)$ is convex and differentiable for all $i \in \mathcal{I}$ with $|f_0^i(x)| = O(|x|^p)$ for some $p \geq 1$.

We use the $\text{SL}(\text{mp})$ property of the supply mechanisms (Assumption 3) to ensure that the value functions are convex, see Lemma 1.

4 Structural Results

In this section, we establish structural results for the optimal dual sourcing policy under various settings. We first consider, in Section 4.1, the case where in each period, the two orders need to be placed in *parallel*. Theorem 1 gives the necessary and sufficient condition for a common threshold policy to be optimal. When there are no salvage opportunities, this condition is always satisfied, see Corollary 1. We also characterize the structure of the optimal policy when the condition fails to prevail.

In Section 4.2, we consider the case where orders can be placed *sequentially*: the regular order can be placed after observing the realization of the expedited order. Theorem 2 establishes that for each supplier, a threshold policy is always optimal. We further show, in Section 4.2.1, that under the special case where the random supply mechanisms consist of random capacities as in (4a), the optimal policies are modified base-stock policies (see Theorem 3 and Corollary 2); however, the policy for the regular supplier sets on a different inventory metric. Section 5 shows that under certain conditions, the expected costs (i.e., the value functions) are pointwise monotone in the state-of-the-world.

4.1 Parallel Orders

We first analyze the model assuming that in each period, orders need to be placed in parallel. The value functions $\{f_n^i(\cdot)\}$ satisfy the recursions:

$$f_n^i(x) = \min_{-C_{n,v}^i \leq q_v \leq 0, \mu_e \in \mathcal{U}_{n,e}^i, \mu_r \in \mathcal{U}_{n,r}^i} \left\{ c_{n,v}^i q_v + c_{n,e}^i \mu_e + c_{n,r}^i \mu_r + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e)) \right. \\ \left. + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j \left(x + q_v + \hat{Y}_{n,e}^i(\mu_e) + \hat{Y}_{n,r}^i(\mu_r) - D_n^i \right) \right\}, \quad (6)$$

where $\mathcal{U}_{n,e}^i = [0, \mathbb{E} \phi_{n,e}^i(C_{n,e}^i, Z_{n,e}^i)]$ and $\mathcal{U}_{n,r}^i = [0, \mathbb{E} \phi_{n,r}^i(C_{n,r}^i, Z_{n,r}^i)]$.

Lemma 1 first establishes among other properties, that the value functions $\{f_n^i(\cdot)\}$ are convex, as long as the terminal value functions $\{f_0^i(\cdot)\}$ are.

Lemma 1 (Parallel orders: convexity property). *(a) The value functions $\{f_n^i(\cdot)\}$ are convex. Moreover, $f_n^i(\cdot)$ is differentiable almost everywhere, with both $|f_n^i(x)|$ and $|f_n^{i'}(x)| = O(|x|^p)$ for some $p \geq 1$, $\lim_{x \downarrow -\infty} f_n^{i'}(x) \leq 0$ and $\lim_{x \uparrow \infty} f_n^{i'}(x) \geq 0$.*

(b) For all $n = N, \dots, 1$ and state of the system (x, i) , there exists a unique triplet of optimal orders and salvage quantity $(q_v^, \mu_e^*, \mu_r^*)$ which optimizes the convex program (6).*

Under *deterministic* supplies, a simple argument proves that it is never optimal to simultaneously place an order with the expedited supplier, and initiate a salvage batch. This also follows as a special case of Theorem 2. Under *random* supply functions, this argument cannot be maintained. It can, however, be guaranteed when the random supply functions $\phi_{n,e}^i(q, Z)$ are supermodular, see Lemma 2 and Remark 4. This property is satisfied for most random supply structures, in particular for the structures in (4a), (4b) and (4c) with $k > -1$, see the Supplemental Material for Federgruen et al. (2023).

Lemma 2 (Parallel orders: no simultaneous salvaging and expedited orders). *Consider problem (6). Assume the underlying random supply functions $\phi_{n,e}^i(q, Z_{n,e}^i)$ are supermodular and increasing in q and $Z_{n,e}^i$. Let $q_v^*(x)$ and $\mu_e^*(x)$ denote the optimal salvage quantity and expedited order in (6), respectively. Then, $\mu_e^*(x) q_v^*(x) = 0$.*

Remark 4. Lemma 2 provides a broad sufficient condition under which placing an expedited order and salvaging are mutually exclusive. This is quite intuitive; indeed, under deterministic supplies, this mutual-exclusivity follows as a special case of Lemma 2, since the supply function $\phi(q, Z) = q$ is supermodular. Even under random supplies, in our extensive numerical studies, we have never found a violation of Lemma 2, but simultaneous expedited ordering and salvaging cannot be excluded without a condition like supermodularity. Indeed, it is conceivable that this phenomenon might occur under random supply functions like the following non-supermodular function.

Let Z be a random variable with a two point distribution having realizations $z_1 = 1$ and $z_2 \gg 1$, with respective probabilities $1 - \epsilon$ and ϵ for some $\epsilon \in (0, 1)$. The random supply function is

$$\phi(q, Z) = q \mathbb{1}\{Z = z_1\} + z_2 \min\{1, q\} \mathbb{1}\{Z = z_2\}.$$

In terms of procurement cost, it of course does not make sense to combine an expedited order with a salvage batch, see Assumption 1. But, given the possible scenario of a very large delivery, one may want to combine the order with a salvage batch to reduce inventory costs.

We are now ready for our first main result (Theorem 1): the *necessary* and *sufficient* condition for the optimal policy to be a common threshold policy; in any given period $n = N, \dots, 1$ and state-of-the-world i , there exist two thresholds $x_{n,0}^i \geq x_{n,1}^i$ such that it is optimal to place an order with one or both suppliers [salvage] when the starting inventory position is below $x_{n,1}^i$ [above $x_{n,0}^i$] and to stay put when the inventory position is in between. Define the following three critical values:

$$x_{n,0}^i := \sup\{x : c_{n,v}^i + L_n^i(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x - D_n^i) < 0\}, \quad (7)$$

$$x_{n,e}^i := \sup\{x : c_{n,e}^i + L_n^i(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x - D_n^i) < 0\}, \quad (8)$$

$$x_{n,r}^i := \sup\{x : c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x - D_n^i) < 0\}, \quad (9)$$

with the convention $\sup \mathbb{R} = \infty$.

The necessary and sufficient condition is

$$x_{n,0}^i \geq x_{n,r}^i. \quad (10)$$

Under Equation (10), the threshold $x_{n,r}^i = \max\{x_{n,r}^i, x_{n,e}^i\}$. Part (b) of Theorem 1 characterizes the structure of our optimal policy when (10) fails to hold.

Theorem 1 (Parallel orders with salvaging: optimality of a common threshold policy). *Fix $n = N, \dots, 1$ and $i \in \mathcal{I}$. Assume the random supply functions $\phi_{n,e}^i(q, Z_{n,e}^i)$ and $\phi_{n,r}^i(q, Z_{n,r}^i)$ are supermodular and increasing in $(q, Z_{n,e}^i)$ and $(q, Z_{n,r}^i)$, respectively.*

(a) *If $x_{n,0}^i \geq x_{n,r}^i$:*

- if $x < \max\{x_{n,r}^i, x_{n,e}^i\}$, place an order, do not salvage, i.e., $q_v^* = 0$, $\mu_e^* + \mu_r^* > 0$;
- if $\max\{x_{n,r}^i, x_{n,e}^i\} < x < x_{n,0}^i$, do nothing, i.e., $q_v^* = \mu_e^* = \mu_r^* = 0$;
- if $x > x_{n,0}^i$, one should only salvage, i.e., $q_v^* < 0$, $\mu_e^* = \mu_r^* = 0$.

In particular, a common threshold policy is optimal.

(b) If $x_{n,0}^i < x_{n,r}^i$:

- if $x < x_{n,0}^i$, it is optimal to not salvage but place an order, i.e., $q_v^* = 0$, $\mu_e^* + \mu_r^* > 0$; or to salvage and only place a regular order, i.e., $q_v^* < 0$, $\mu_r^* > 0$, $\mu_e^* = 0$.
- if $x_{n,0}^i < x < x_{n,r}^i$, it is optimal to salvage and only place an order with the regular supplier, i.e., $q_v^* < 0$ and $\mu_r^* > 0$, $\mu_e^* = 0$;
- if $x > x_{n,r}^i$, it is optimal to salvage and not place an order with the expedited supplier, i.e., $q_v^* < 0$, $\mu_e^* = 0$.

In particular, no common threshold is optimal.

In the absence of salvage opportunities, the existence of an optimal threshold policy is guaranteed in *all* settings. Note that when the salvage revenue rates $\{c_{n,v}^i\} \downarrow -\infty$, the values $\{x_{n,0}^i\} \uparrow \infty$ and the necessary and sufficient condition in Theorem 1 is trivially satisfied. Moreover, supermodularity of the supply mechanism is only necessary in Lemma 2 to ensure $\mu_e^* q_v^* = 0$, but since $q_v^* = 0$ (when salvaging is not permitted), this complementarity condition is trivially satisfied.

Corollary 1 (Parallel orders without salvaging). *Consider the model with no salvage options. Then for every period n and state-of-the-world i there exists a critical threshold x^* for the inventory position such that it is optimal to place an order with one or both suppliers iff $x \leq x^*$.*

Indeed, in Theorem 1, the case where $x_{n,0}^i < x_{n,r}^i$ is rare so that the optimality of a common threshold policy generally applies (see part (a)). However, this is the case in the following example.

Example 1. Consider an un-modulated instance so that the superscript i can be dropped. Let $N = 5$, $\alpha = 1$ and $h_n = 1$, $p_n = 5$ for $n = N, \dots, 1$, while $l_e = 0$, $l_r = 1$. Assume for $n = N, \dots, 1$, $c_{n,e} = 3$, $c_{n,v} = 1.5$ and $c_{n,r} = 2$ with corresponding capacity levels $C_{n,e} = 2$, $C_{n,v} = C_{n,r} = 5$. The demand distribution is uniform on $\{0, \dots, 5\}$ and all supplies are deterministic.

In this case $5 = x_{4,0} < x_{4,r} = 6$. Figure 1 exhibits the optimal strategy, with the blue dots representing the inventory position after salvaging and a parallel order with the expedited supplier. The yellow dots show where that (adjusted) inventory position then reaches through an order with the regular supplier.

Note that on the interval $(5, 13)$ the blue dots lie below the 45° line, indicating that part of the inventory is being salvaged: $q_v^* < 0$. By Lemma 2 (which applies trivially to deterministic supplies), this implies that $\mu_e^* = 0$. However, on this interval, the blue dots are strictly below the yellow dots, implying that an order with the regular supplier accompanies the salvage batch, i.e., $\mu_r^* > 0$ while $q_v^* < 0$, thus precluding the optimality of a common threshold policy.

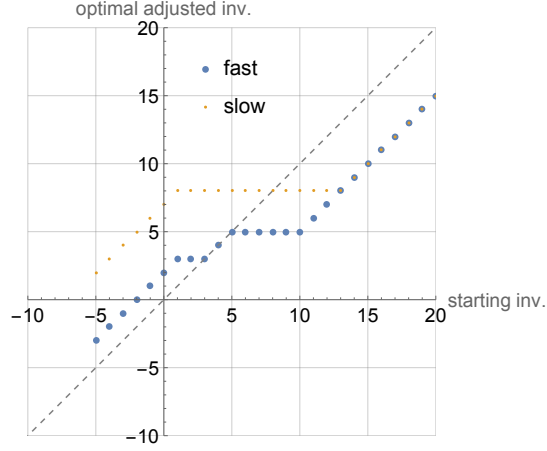


Figure 1: An example for $x_{n,0}^i < x_{n,r}^i$ (period $n = 4$)

4.2 Sequential Orders

We now address settings where the orders to the different suppliers may be placed *sequentially*, such that the realization of the random supply generated by an earlier order becomes known before any later order need to be placed. This assumption can be made, without loss of optimality, under deterministic supplies, since there is only one realization of any order. Beyond deterministic cases, such sequential ordering is common in manufacturing (e.g. semiconductor), where buyers observe the actual yield rate from a primary supplier’s production run before determining orders from backup suppliers.

Specifically, we assume, in any given period, the realization of the random supply generated by the expedited order becomes known *before* an order to the regular supplier needs to be placed. The realization of the random supply generated by the regular order is then observed in the same period, even though its delivery requires a lead time l_r . A variant of this model, where one first places an order with the regular supplier, and upon learning its realization, potentially supplemented with an expedited order, is developed in Appendix C. Clearly, the ability to use this information is valuable and results in lower expected costs. Moreover, we show that, in this setting, the optimal policies have additional structural properties which the optimal policy under *parallel* ordering may fail to possess. In particular, an individual threshold policy is always optimal for the ordering policy of *both* suppliers. The optimal policy for the regular supplier sets on the *adjusted inventory position*, defined as the initial inventory position after inclusion of an order with the expedited supplier or salvage quantity. In the special case of a “random capacity model” (see Section 4.2.1), one can show that the optimal policies are (modified) base-stock policies.

Thus, let $f_{n,r}^i(y)$ be the minimum total present value of all costs in the last n periods, assuming the state-of-the-world in period n is i , and that any order with the expedited supplier or salvage quantity determined at the start of period n has resulted in an *adjusted* inventory position of y units. Then the

value function is given by

$$f_n^i(x) = \min_{-C_{n,v}^i \leq q_v \leq 0, \mu_e \in \mathcal{U}_{n,e}^i} \left\{ c_{n,v}^i q_v + c_{n,e}^i \mu_e + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e)) + \mathbb{E} f_{n,r}^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e)) \right\}, \quad (11)$$

where

$$f_{n,r}^i(y) = \min_{\mu_r \in \mathcal{U}_{n,r}^i} \left\{ c_{n,r}^i \mu_r + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(y + \hat{Y}_{n,r}^i(\mu_r) - D_n^i) \right\}. \quad (12)$$

Theorem 2 (Sequential orders with salvaging: optimality of individual threshold policies). *Consider the general model with sequential ordering, described by the dynamic program recursion (11) and (12).*

- (a) *The value functions $f_n^i(\cdot)$ and $f_{n,r}^i(\cdot)$ are convex, and hence almost everywhere differentiable with $|f_n^{i'}(x)| = O(|x|^p)$ and $|f_{n,r}^{i'}(y)| = O(|y|^p)$ for some $p \geq 1$, with $\lim_{x \downarrow -\infty} f_n^{i'}(x) < 0$, $\lim_{x \uparrow \infty} f_n^{i'}(x) > 0$ and $\lim_{y \downarrow -\infty} f_{n,r}^{i'}(y) < 0$, $\lim_{y \uparrow \infty} f_{n,r}^{i'}(y) > 0$.*
- (b) *Fix a period $n = N, \dots, 1$ and state-of-the-world $i \in \mathcal{I}$. Let $\underline{x} = x + q_v$ denote the starting inventory position after a possible salvage reduction, and $\underline{x}^*$ its optimal value. The optimal expedited order target $\mu_e^*(x) \geq 0$ is non-increasing in x , while $\underline{x}^*$ is non-decreasing in x . Similarly, the optimal order target $\mu_r^*(y) \geq 0$ for the regular supplier is non-increasing in y .*
- (c) *Assume, in addition, that the random supply functions $\phi_{n,e}^i(q, Z)$ and $\phi_{n,r}^i(q, Z)$ are supermodular, and increasing in Z (as well as in q). Fix a period $n = N, \dots, 1$. The optimal first-stage policy is an individual threshold policy, i.e., there exist two thresholds x_1^*, x_2^* such that it is optimal to place only an expedited order when $x < x_1^*$, to only salvage when $x > x_2^*$, and to do neither when $x_1^* \leq x \leq x_2^*$. Similarly, there exists a threshold y^* , such that it is optimal to place a regular order iff the adjusted inventory position y (after inclusion of an order with the expedited supplier or salvage quantity) has $y < y^*$. (All thresholds are period dependent and dependent on the state-of-the-world.)*

4.2.1 Special Case: Random Capacity Model

We now turn our attention to the *first* type of random supply functions in (4a), i.e., the case where the supplier's capacity level is random, once again assuming that orders can be placed sequentially. Under this very important random supply structure, the optimality result of threshold policies in Theorem 2 can be sharpened: each of the order policies as well as the salvaging strategy are (modified) base-stock policies, in every period and for any given state-of-the-world at the start of the period.

Indeed, in the first stage, there are two thresholds $x_1^* \leq x_2^*$ such that it is optimal to elevate [reduce] the inventory position to come as close to x_1^* [x_2^*] as possible, with an expedited supplier order [salvage batch], when the starting inventory position $x \leq x_1^*$ [$x \geq x_2^*$]. Similarly, in the second stage, it is optimal to elevate the *adjusted inventory position* to come as close as possible to y^* . Thus,

$q_e^* = (x_1^* - x)^+$, $q_v^* = \min\{x_2^* - x, 0\}$ and $q_r^* = (y^* - y)^+$, with the understanding that these target interventions are capped by the relevant random capacity levels.

We represent the expedited [regular] supplier's capacity in period n under a state-of-the-world $i \in \mathcal{I}$ as $C_{n,e}^i + \xi_{n,e}^i [C_{n,r}^i + \xi_{n,r}^i]$, with $C_{n,e}^i$ and $C_{n,r}^i$ the *expected* capacity levels, while the random noise terms $\xi_{n,e}^i, \xi_{n,r}^i$ have $\mathbb{E}\xi_{n,e}^i = \mathbb{E}\xi_{n,r}^i = 0$ and convex supports $\mathcal{X}_{n,e}^i \subset [-C_{n,e}^i, \infty)$, $\mathcal{X}_{n,r}^i \subset [-C_{n,r}^i, \infty)$. Assume, further, that the triplet $\{\xi_{n,e}^i, \xi_{n,r}^i, D_n^i\}$ are independent, for each $n = N, \dots, 1$ and $i \in \mathcal{I}$.

The value functions now satisfy the following dynamic program recursions:

$$f_n^i(x) = \min_{q_e \geq 0, -C_{n,v}^i \leq q_v \leq 0} \mathbb{E} [c_{n,v}^i q_v + c_{n,e}^i (q_e \wedge (C_{n,e}^i + \xi_{n,e}^i)) + g_n^i(x + q_v + q_e \wedge (C_{n,e}^i + \xi_{n,e}^i))],$$

where $g_n^i(y) = L_n^i(y) + f_{n,r}^i(y)$ and

$$f_{n,r}^i(y) = \min_{q_r \geq 0} \mathbb{E} \left[c_{n,r}^i (q_r \wedge (C_{n,r}^i + \xi_{n,r}^i)) + \alpha \sum_{j=1}^I p_{ij} f_{n-1}^j(y + q_r \wedge (C_{n,r}^i + \xi_{n,r}^i) - D_n^i) \right].$$

Let $y = x + q_e$ and $z = y + q_r$. We can rewrite the above problem as

$$f_n^i(x) = \min_{y \geq x, -C_{n,v}^i \leq q_v \leq 0} \mathbb{E} [c_{n,v}^i q_v + c_{n,e}^i (y \wedge (x + C_{n,e}^i + \xi_{n,e}^i) - x) + g_n^i(y \wedge (x + C_{n,e}^i + \xi_{n,e}^i) + q_v)],$$

and

$$f_{n,r}^i(y) = \min_{z \geq y} \mathbb{E} \left[c_{n,r}^i (z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) - y) + \alpha \sum_{j=1}^I p_{ij} f_{n-1}^j(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) - D_n^i) \right].$$

Define sets

$$\begin{aligned} \mathcal{A}_{n,e}^i(x) &= \{v : v = y \wedge (x + C_{n,e}^i + \xi_{n,e}^i), y \geq x, \xi_{n,e}^i \in \mathcal{X}_{n,e}^i\}, \\ \mathcal{A}_{n,r}^i(y) &= \{v : v = z \wedge (y + C_{n,r}^i + \xi_{n,r}^i), z \geq y, \xi_{n,r}^i \in \mathcal{X}_{n,r}^i\}. \end{aligned}$$

Lemma 3. $\mathcal{A}_{n,e}^i(x) = [x, \bar{y}(x)]$ and $\mathcal{A}_{n,r}^i(y) = [y, \bar{z}(y)]$ for some $\bar{y}(x) \leq \infty$ and $\bar{z}(y) \leq \infty$.

The next theorem and corollary establish that the optimal policy is a combination of *modified base-stock policies*. As with Theorem 1/Corollary 1, the structure further simplifies in the absence of salvaging opportunities.

Theorem 3 (Random capacities with salvaging: optimality of modified base-stock policies). *Consider the random capacity model with sequential ordering. Fix a period $n = N, \dots, 1$. The optimal first-stage policy is a double modified base-stock policy with two thresholds $x_1^* \leq x_2^*$ so that it is optimal to place an expedited order $q_e^* = x_1^* - x$ when $x < x_1^*$, to initiate a salvage batch $q_v^* = x_2^* - x$ when $x > x_2^*$, and to do neither when $x_1^* \leq x \leq x_2^*$. Similarly, the second-stage policy is a modified base-stock policy with a threshold y^* so that it is optimal to place a regular order $q_r^* = (y^* - y)^+$. The threshold values depend on the period index n and the state-of-the-world i . The optimal order policies are clearly decreasing in x and the optimal salvaging policy is increasing in x .*

Remark 5. Note that, in any period $n = N, \dots, 1$, the optimal base-stock levels $\{z_{n,i}^*\}$ and $\{y_{n,i}^*\}$ are entirely independent of that period's capacity distributions. Through the shape of the value functions $\{f_{n-1}^j(\cdot)\}$ they do depend on all future capacity distributions.

The following corollary is immediate from Theorem 3 and its proof.

Corollary 2 (Random capacities without salvaging). *Consider the random capacity model with sequential ordering and no salvage options. Let $y_{n,i}^*$ denote the minimizer of function $c_{n,e}^i y + g_n^i(y)$, and $z_{n,i}^*$ the minimizer of function $c_{n,r}^i z + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(z - D_n^i)$. The optimal expedited order size is $(y_{n,i}^* - x)^+$ when x is the starting inventory position in period n , under state-of-the-world i . Based on the realized adjusted inventory position y , it is optimal to place a regular order of size $(z_{n,i}^* - y)^+$.*

Remark 6. The optimality of a modified base-stock policy for each of the suppliers was already shown in Yazlali and Erhun (2007) for the special case of deterministic capacities. In Corollary 2, we extend this result to general random supply mechanisms. Tan et al. (2016) addressed the special case of this model, where $l_e = 1, l_r = 2$, holding and backlogging costs are linear, there are no salvage opportunities and there is no Markov modulation of the parameters or the demand distributions, albeit under parallel ordering. The authors claim the optimality of a threshold policy for each supplier in their Proposition 1, but this claim is not substantiated in their proof or elsewhere, see Online Appendix OA.2. The example in Figure 2 on p.1237 of Tan et al. (2016) shows that a modified base-stock policy may fail to be optimal. However, contrary to what is stated there, this is not due to the randomness of the supply mechanisms (see Corollary 2) but due to orders being placed in parallel (see Theorem 1).

5 Monotonicity of Value Functions w.r.t. the State-of-the-World

In this section, we show that under some conditions the value functions are monotone in the state-of-the-world. We show these results for the case of parallel ordering; similar results can be obtained under sequential ordering. We first assume that all model parameters are independent of the state-of-the-world except for the cost rates. The cost rates are all ascending in the state-of-the-world, i.e., for all $n = N, \dots, 1$:

$$c_{n,r}^1 \leq \dots \leq c_{n,r}^I; \quad c_{n,e}^1 \leq \dots \leq c_{n,e}^I; \quad c_{n,v}^1 \leq \dots \leq c_{n,v}^I. \quad (13)$$

If only these cost rates depend on the state-of-the-world, one would expect that all optimal total expected costs are increasing in the initial state-of-the-world, for any starting period n and starting inventory position x . This can, indeed, be guaranteed as long as the state-of-the-world in the next period, i_{n-1} ,

is stochastically increasing in the state-of-the-world in the current period, i_n , i.e.,

$$(i_{n-1}|i_n = i) \leq_{st} (i_{n-1}|i_n = j), \quad \text{for any pair } 1 \leq i < j \leq I. \quad (14)$$

See [Massey \(1987\)](#) for a general treatment of stochastically monotone Markov processes. [Song and Zipkin \(1993\)](#) used the same stochastic monotonicity condition to prove that the optimal policy thresholds are monotone in the state-of-the-world. In their single sourcing model without capacity limits and salvage opportunities, the demand process is Poisson with a rate depending on the state-of-the-world.

Theorem 4 (Optimal cost monotone in state-of-the-world under price fluctuations). *Consider a general Markov-modulated dual sourcing model, in which only the unit cost rates $\{c_{n,r}^i\}, \{c_{n,e}^i\}, \{c_{n,v}^i\}$ depend on the state-of-the-world. Assume conditions (13) and (14) apply and the terminal value function $f_0^j(\cdot)$ is pointwise increasing in j . Then $f_n^j(x)$ is increasing in j for all $n = N, \dots, 1$ and $x \in \mathbb{R}$.*

Now consider a Markov chain that affects the two suppliers' capacity levels. As before, we assume the states-of-the-world $\mathcal{I} = \{1, \dots, I\}$ are ranked in decreasing order of attractiveness, which, in terms of capacity levels, means:

$$\begin{aligned} \mathbb{E}[\phi_{n,r}^1(C_{n,r}^1, Z_{n,r}^1)] &\geq \dots \geq \mathbb{E}[\phi_{n,r}^I(C_{n,r}^I, Z_{n,r}^I)], \\ \mathbb{E}[\phi_{n,e}^1(C_{n,e}^1, Z_{n,e}^1)] &\geq \dots \geq \mathbb{E}[\phi_{n,e}^I(C_{n,e}^I, Z_{n,e}^I)]. \end{aligned} \quad (15)$$

Theorem 5 establishes that the value functions are, correspondingly, pointwise increasing in the state-of-the-world, assuming the state-of-the-world in the next period is stochastically increasing in the current state-of-the-world as in (14).

Theorem 5 (Optimal cost monotone in state-of-the-world under capacity fluctuations). *Assume the state-of-the-world affects the suppliers' capacities, as in (15). Condition (14) applies and the terminal value function $f_0^j(\cdot)$ is pointwise increasing in j . Also assume that the random supply functions satisfy $\hat{Y}_{n,r}^j(\mu) \geq_{\text{convex}} \hat{Y}_{n,r}^i(\mu)$ and $\hat{Y}_{n,e}^j(\mu) \geq_{\text{convex}} \hat{Y}_{n,e}^i(\mu)$ for any $1 \leq i \leq j \leq I$. Then $f_n^j(x)$ is increasing in j for all $n = N, \dots, 1$ and $x \in \mathbb{R}$.*

6 Two Types of Volatilities: Their Impact on Costs

Until now, we have focused on how ordering and salvaging decisions depend on the state of the system. In this section, we focus on how different types of volatilities impact the expected costs.

In our model, we recognize two types of volatilities. There are those that emerge from “internal factors”, in particular, the demand distributions and supply mechanisms (*internal volatilities*). In addition, there may be volatility because various model primitives, such as the cost rates, capacity levels, demand distributions, supply and $L_n(\cdot)$ functions, may all depend on an “external” state-of-the-world, which fluctuates according to an exogenous Markov chain (*external volatilities*).

6.1 Internal Volatilities

As mentioned in the introduction, the two types of volatilities have *different* impacts on the costs in dual or multi-supplier settings. While an increase of internal volatilities typically results in *increased* costs, the opposite may prevail with respect to *external* volatilities. The difference arises because the realization of internal volatilities from random demands or supplies cannot be observed in advance of the relevant ordering decisions, while the fluctuating state-of-the-world is observable *before* these inventory decisions need to be made: a dual sourcing supply chain allows one to react to the changing state-of-the-world by shifting orders between the suppliers (as well as adjusting aggregate orders).

It is well known, since the inception of inventory theory, that an increase in demand volatility, due to a more volatile demand distribution, typically results in an increase of safety stocks and total costs. This is in contrast to the dependence of the demand distribution on the state-of-the-world. With this information known a priori, it can be exploited when making ordering decisions. The expected cost increase is apparent under Normal demands, from the newsvendor formula or that of the optimal base-stock level in multi-period models with stationary demands, which are both affine in the standard derivation of the demand distribution.

Perhaps the most popular notion for comparing the variability of two random variables is the convex order $\geq_{convex}$. It is immediate that

$$X \geq_{convex} Y \implies \mathbb{E}X = \mathbb{E}Y \text{ and } \text{Var}(X) \geq \text{Var}(Y). \quad (16)$$

Moreover, there exists a random variable ϵ with $\mathbb{E}\epsilon = 0$ and $\mathbb{E}[\epsilon|x] \geq 0$ almost surely such that $X = Y + \epsilon$ in distribution, see [Brumelle and Vickson \(1975\)](#). Therefore X is noisier than Y .

In the continuous review model with Poisson demand governed by an (r, q) -policy, Theorem 5 of [Song et al. \(2010\)](#) shows that the long-run average cost is larger under any (r, q) -policy—and, a fortiori, under the optimal (r, q) -policy—when the lead time distribution is replaced by a convexly larger one. [Federgruen and Wang \(2013\)](#) generalize this result for a general system governed by a (r, q) -policy; the authors show that the optimal average cost *increases* when the lead-time demand distribution is replaced by a convexly larger one, see Theorem 6 and the subsequent discussion.

Similarly, all expected costs increase when any of the supply mechanisms is replaced by a convexly-ordered larger one. (Recall this means that the mean remains constant but the variance increases, see (16)). Thus, compare two systems A and B that are identical except for their supply mechanisms: for all n, i, μ_e and μ_r ,

$$\hat{Y}_{n,e}^{i,B}(\mu_e) \geq_{convex} \hat{Y}_{n,e}^{i,A}(\mu_e) \text{ and } \hat{Y}_{n,r}^{i,B}(\mu_r) \geq_{convex} \hat{Y}_{n,r}^{i,A}(\mu_r). \quad (17)$$

Let $f_{n,A}^i(\cdot)$ and $f_{n,B}^i(\cdot)$ denote their value functions, with a common terminal value function $f_0(\cdot)$. For the sake of brevity, we confine ourselves to the case of parallel ordering.

Theorem 6 (Increased Costs due to More Volatile Supply Mechanisms). *Consider systems A and B above with random supply mechanisms that satisfy (17) and assume parallel ordering. Then for all n, i ,*

$$f_{n,B}^i(x) \geq f_{n,A}^i(x), \quad \forall x.$$

As a special case of Theorem 6, we obtain from Jensen's inequality that deterministic supplies are always cheaper than any kind of random supplies with identical means.

6.2 External Volatilities

In this subsection, we demonstrate that the impact of external volatilities can be quite different from those discussed above, for internal volatilities. We derive various comparison results between pairs of systems that are differentiated only in terms of the specific way one of the model parameters fluctuates with the state-of-the-world. We focus, in particular, on the unit procurement cost rates $\{c_{n,e}^i\}$ and $\{c_{n,r}^i\}$ as well as the capacity levels $\{C_{n,e}^i\}$ and $\{C_{n,r}^i\}$.

To begin with, consider settings where the unit cost rates for the regular supplier are *independent* of the state-of-the-world, while those pertaining to the expedited supplier are *state-dependent*. The results for the reverse case are analogous. Our first proposition (Proposition 1) compares a world where the dependence of the cost rate of the expedited supplier $\{c_{n,e}^i\}$, with respect to the state-of-the-world i , is minimal, in the limit non-existent, versus one where there is significant fluctuation of $\{c_{n,e}^i\}$ as a function of i . To compare apples with apples, we assume that in the second system, the *expected* unit cost rate in every period is equal to (or lower than) the deterministic cost rate in the first system.

Consider therefore two systems in a general model. In system A (the *stable price* world), all unit cost rates $\{c_{n,e}^i\}$ are independent of the state-of-the-world $\{i_n\}$ progression, i.e., for $n = N, \dots, 1$, $c_{n,e}^i \equiv c_{n,e}^*$, $\forall i \in \mathcal{I}$, where $c_{n,e}^*$ is a constant. In system B (the *volatile price* world), the cost rates $\{c_{n,e}^i\}$ have a general dependence on the state-of-the-world, however with $\mathbb{E}[c_{n,e}^{i_n} | c_{N,e}^{i_N} = c_{N,e}^*] \leq c_{n,e}^*$, $n = N, \dots, 1$. Assume, further, that only $\{c_{n,e}^i\}$ are dependent on the state-of-the-world but not the remaining model primitives, in particular the unit cost rates of the regular supplier, the demand distributions, the random supply functions, the $L_n(\cdot)$ functions, or the capacity levels. In this case, we obtain a strong comparison result between the two systems: under *any* demand trajectory, expected total costs are lower in the volatile price world (system B).

Proposition 1 (Stable price world v.s. volatile price world). *Consider systems A and B as defined above. In particular, $\mathbb{E}[c_{n,e}^{i_n} | c_{N,e}^{i_N} = c_{N,e}^*] \leq c_{n,e}^*$ for $n = N, \dots, 1$.*

- (a) *Fix an arbitrary demand trajectory $\{d_n\}$. The expected optimal total costs under the volatile price model is lower than that under the stable price model, for any starting period n and state of the system (x, i) .*

(b) In particular, let $\{f_{n,A}^i(\cdot)\}$ and $\{f_{n,B}^i(\cdot)\}$ denote the value functions in systems A and B, respectively. $f_{n,B}^i(x) \leq f_{n,A}^i(x)$ for all $n = N, \dots, 1$, $i \in \mathcal{I}$ and $x \in \mathbb{R}$.

The proof shows that the optimal policy π in system A incurs a lower expected cost in system B, under *any* demand trajectory. Clearly, policy π can be improved in system B in several ways: *first*, if the expected unit cost rate for the expedited supplier is $c_{n,e}^*$, there is a set of states-of-the-world $\mathcal{I}^-$ in which $c_{n,e}^i < c_{n,e}^*$ and a set $\mathcal{I}^+$ where $c_{n,e}^i > c_{n,e}^*$. Even without altering the *total* orders placed on any given sample path, total costs can be lowered by assigning a larger [smaller] part of the order to the expedited supplier when the state-of-the-world is in $\mathcal{I}^-$ [$\mathcal{I}^+$], and the magnitude of the desired shifts depends, quite intuitively, on the magnitude of $c_{n,e}^* - c_{n,e}^i$. *Second*, when encountering a favorable [unfavorable] state-of-the-world $i_n \in \mathcal{I}^-$ [$\mathcal{I}^+$], it is, usually, beneficial to increase [reduce] the *total* order size as well. (See the discussion following Proposition 2.)

Note that the condition $\mathbb{E}[c_{n,e}^{i_n} | c_{N,e}^{i_N} = c_{n,e}^*] \leq c_{n,e}^*$ for any n can be satisfied by various types of Markov chains, for example:

Example 2. Consider a Markov chain where $\mathbb{E}[c_{n,e}^{i_n} | c_{n+1,e}^{i_{n+1}}] \leq c_{n,e}^*$ for any value of i_{n+1} . Then we have $\mathbb{E}[c_{n,e}^{i_n} | c_{N,e}^{i_N}] \leq c_{n,e}^*$.

Example 3. Assume $\{c_{n,e}^*\}$ is stationary, i.e., $c_{n,e}^* \equiv c_e^*$, where c_e^* is a constant. Consider a Markov chain where $\{c_{n,e}^{i_n} : n = N, \dots, 1\}$ is a submartingale, so that $\mathbb{E}[c_{n,e}^{i_n} | c_{n+1,e}^{i_{n+1}}] \leq c_{n+1,e}^{i_{n+1}}$. Then starting from $c_{N,e}^{i_N} = c_e^*$, we have $\mathbb{E}[c_{n,e}^{i_n} | c_{N,e}^{i_N} = c_e^*] \leq c_e^*$.

We generalize Proposition 1 by comparing *two volatile* price worlds, with the same Markov chain determining the evolution of the state-of-the-world $\{i_n, n = N, \dots, 1\}$. The two systems differ only in the dependence of the unit cost rates (for the expedited supplier) on the state-of-the-world. Variations in these cost rates, as a function of the state-of-the-world, are less severe in the first system A as compared to the second one B. More specifically, let $\{c_{n,e}^i\}$ and $\{\tilde{c}_{n,e}^i\}$ denote the set of cost rates in the two systems, and assume, without loss of generality, that cost rates are ascending in the state-of-the-world:

$$c_{n,e}^1 \leq c_{n,e}^2 \leq \dots \leq c_{n,e}^I \quad \text{and} \quad \tilde{c}_{n,e}^1 \leq \tilde{c}_{n,e}^2 \leq \dots \leq \tilde{c}_{n,e}^I. \quad (18)$$

Furthermore, assume there exists a state $1 \leq k \leq I$ such that for all $n = N, \dots, 1$:

$$\tilde{c}_{n,e}^i \leq c_{n,e}^i, \text{ for } i = 1, \dots, k, \quad \text{and} \quad \tilde{c}_{n,e}^i \geq c_{n,e}^i, \text{ for } i = k+1, \dots, I. \quad (19)$$

Thus, the cost rates in the second, more volatile system span a larger spectrum of values.

Remark 7. Assumption (19) stipulates a single crossing-state k when comparing the price vectors $\{c_{n,e}^i\}$ and $\{\tilde{c}_{n,e}^i\}$. It follows from the proof of Proposition 2 that this crossing-state k may be time-dependent, i.e., $k = k_n, n = N, \dots, 1$.

An additional assumption is that, in any given period and state-of-the-world, the expected cost rate

in the *next* period is at least as large and therefore as expensive, see (18), in the first, more stable system as compared to the second, more volatile system: for any given identical state-of-the-world $i \in \mathcal{I}$ and $n = N, \dots, 1$,

$$\sum_{j=1}^I p_{ij} c_{n-1,e}^j = \mathbb{E}[c_{n-1,e}^{i_{n-1}} | i_n = i] \geq \mathbb{E}[\tilde{c}_{n-1,e}^{i_{n-1}} | i_n = i] = \sum_{j=1}^I p_{ij} \tilde{c}_{n-1,e}^j. \quad (20)$$

Proposition 2 below shows that under Conditions (19) and (20), all cost-to-go functions are lower in the more volatile system. In particular, it is interesting that we merely need to guarantee that the expected future procurement costs are lower in the more volatile system for a given starting state, irrespective of how much more volatile these procurement costs may be. For example, see the numerical examples in Table 2 in Section 6.3.

Unlike Proposition 1, we allow the state-of-the-world to impact capacity levels, demand distributions, the supply mechanisms and $L_n^i(\cdot)$ functions, the unit cost rates of both suppliers. To isolate the impact of the latter cost rates, we merely assume the dependence of all other model primitives on the state-of-the-world, as identical in both systems A and B.

Proposition 2 (Benefiting from increased price volatility). *Consider the “more stable price model”, system A, and the “more volatile price model”, system B, defined above. In particular, assume conditions (18)–(20) apply. Let $\{f_n^i(\cdot)\}$ and $\{\tilde{f}_n^i(\cdot)\}$ denote the value functions in systems A and B, respectively. Let $\mu_{n,e}^*(x, i)$ denote the optimal targeted order size with the expedited supplier, in system A, in period n when starting in state (x, i) . Assume $\{\mu_{n,e}^*(x, i)\}$ is decreasing in i . If $\tilde{f}_0^j(\cdot) = f_0^j(\cdot)$ for all $j \in \mathcal{I}$, then $\tilde{f}_n^i(x) \leq f_n^i(x)$ for all $n = N, \dots, 1$, $i = 1, \dots, k$ and $x \in \mathbb{R}$.*

Proposition 2 shows that a *more volatile* price world may be exploited to the decision maker’s advantage. This is in contrast to the internal volatilities described above. The fundamental difference is that, in our model the decision maker is able to observe the state-of-the-world and exploit it to adjust her ordering pattern. In particular, it allows the decision maker to increase her order with the expedited supplier in low-indexed states where, in the more volatile price model, the prices charged by the expedited supplier go down. Conversely, it allows the decision maker to reduce the order sizes to the expected supplier in high-indexed states where the unit prices increase relative to those in the more stable price world (system A).

Remark 8. *Proposition 2 requires the optimal order sizes $\{\mu_{n,e}^*(x, j)\}$ or targeted quantities $\{\mu_{n,e}^*(x, j)\}$ to be monotonically decreasing in the state-of-the-world j .*

This monotonicity property is intuitive and holds with great generality. However, it is sometimes violated, as the following example with deterministic supplies shows: consider a stationary model without salvage options and with deterministic supplies, modulated by a 3-state Markov chain: the *good*, *middle* and the *bad* state as manifested by the expedited supplier’s unit cost rates $c_{n,e}^1 = 15$, $c_{n,e}^2 = 20$ and $c_{n,e}^3 = 80$. We normalize the regular supplier’s unit cost rate $c_{n,r} = 0$, in every state-of-the-world.

The regular [expedited] supplier has a capacity level $C_r = 6$ [$C_e = 3$]. The single period demand is normally distributed with mean $\mu = 5$ and standard deviation $\sigma = 2$. Unit holding and backlogging cost rates are $h = 5$ and $p = 95$, respectively. The lead times are $l_e = 2$ and $l_r = 3$. The transition probability matrix is

$$P = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

In Figure 2, below, we display the order quantities $q_{n,e}^*(x, i)$ as prescribed by the optimal policy for a remaining horizon $n \geq 36$. Note that when the starting inventory level x is in the range $[16, 19]$, a *larger* order is placed with the expedited supplier, in the middle state as compared to the good state, a violation of the monotonicity assumption in Proposition 2.

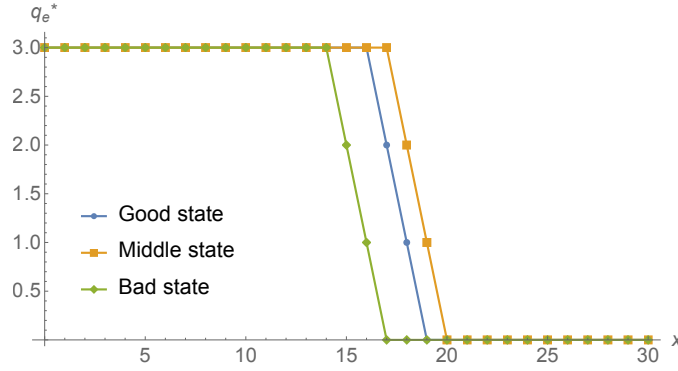


Figure 2: An example violating the monotonicity assumption in Proposition 2

The intuition behind this at first surprising phenomenon is as follows. When we are in the good state, we remain there for a geometrically distributed number of periods, on average $(0.2)^{-1} = 5$ periods. But as soon as we land in the middle state, we move in one single period to the bad state to remain there forever. Thus, when arriving in the middle state, it pays to *increase* the order with the expedited supplier (compared to that placed in the good state) to reduce future cost surges.

However, numerical experiments indicate that violation of the monotonicity of $\{q_{n,e}^*(x, i)\}$ in i is very rare. As a consequence, this monotonicity property can be imposed as an *upfront* restriction to the class of admissible policies, without significant or any loss of optimality. Furthermore, these numerical experiments show that even if the monotonicity assumption on $\{q_{n,e}^*(x, i)\}$ is violated, the conclusion in Proposition 2 typically *still* holds.

We now compare two Markov-modulated dual sourcing models, with $\mathcal{I} = \{1, \dots, I\}$ a common set of states-of-the-world, and with identical cost parameters, capacity levels, and random supply mechanisms and demand distributions, but different Markov processes for the state-of-the-world. Let $\{i_n\}$ and $\{\tilde{i}_n\}$ denote these Markov processes in system A and B, respectively. As before, we assume that the capacity limits $\{C_{n,r}^i\}, \{C_{n,e}^i\}$ and $\{C_{n,v}^i\}$, demand distributions, supply mechanisms and $L_n^i(\cdot)$ functions are

independent of the state-of-the-world. Only, the procurement and salvage cost rates are increasing in the state-of-the-world. Moreover, in both systems the state-of-the-world in the next period is stochastically increasing in the current state-of-the-world, i.e., for any pair of states $1 \leq i < j \leq I$ and all $n = N, \dots, 1$:

$$(i_{n-1}|i_n = i) \leq_{st} (i_{n-1}|i_n = j), \quad (\tilde{i}_{n-1}|\tilde{i}_n = i) \leq_{st} (\tilde{i}_{n-1}|\tilde{i}_n = j). \quad (21)$$

Also, for any state-of-the-world in period n , the state-of-the-world in the next period is stochastically larger under system A versus system B:

$$(i_{n-1}|i_n = j) \geq_{st} (\tilde{i}_{n-1}|\tilde{i}_n = j) \quad n = N, \dots, 1, j \in \mathcal{I}. \quad (22)$$

Proposition 3 (Comparing different Markov processes under price fluctuations). *Consider two general Markov-modulated dual sourcing models with a common set of states-of-the-world $\mathcal{I} = \{1, \dots, I\}$ and identical cost parameters, capacity limits and demand distributions. Moreover, all random supply functions and demand distributions are independent of the state-of-the-world. Assume all unit procurement cost rates are increasing in the state-of-the-world, as in (13), and the Markov processes $\{i_n\}$ and $\{\tilde{i}_n\}$ satisfy (21) and (22). Let $f_n^i(\cdot)$ and $\tilde{f}_n^i(\cdot)$ denote the value functions in systems A and B, respectively. Then, $f_n^i(x) \geq \tilde{f}_n^i(x)$ for all $n = N, \dots, 1$, $i \in \mathcal{I}$ and $x \in \mathbb{R}$, if $f_0^i(\cdot) \geq \tilde{f}_0^i(\cdot)$ for all $i \in \mathcal{I}$.*

In Proposition 3, only the unit cost rates vary with the state-of-the-world, and these rates increase with the latter. Proposition 4 provides the direct counterpart of this proposition, where only the capacity levels of the two suppliers vary with the state-of-the-world, and in this case, decreasingly so, as in (15). While in Proposition 3, a higher indexed state-of-the-world implies higher unit cost rates, in Proposition 4 it implies lower, i.e., less attractive, capacity levels. The two systems are again driven by two distinct Markov chains. As in Proposition 3, we assume that under both Markov chains, the next period's state-of-the-world is stochastically increasing in the current state, see (21). Moreover, for any current state-of-the-world, the next period's state-of-the-world is stochastically larger under system A, as compared to system B, see (22).

Proposition 4 (Comparing different Markov processes under capacity fluctuations). *Consider two general Markov-modulated dual sourcing models with a common set of states-of-the-world $\mathcal{I} = \{1, \dots, I\}$. Assume the capacity limits for the two suppliers are decreasing in the state-of-the-world, as in (15); all other model parameters are independent of the state-of-the-world. The Markov processes $\{i_n\}$ and $\{\tilde{i}_n\}$ satisfy (21) and (22). Let $f_n^i(\cdot)$ and $\tilde{f}_n^i(\cdot)$ denote the value functions in systems A and B, respectively. Then, $f_n^i(x) \geq \tilde{f}_n^i(x)$ for all $n = N, \dots, 1$, $i \in \mathcal{I}$ and $x \in \mathbb{R}$, provided $f_0^i(\cdot) \geq \tilde{f}_0^i(\cdot)$ for all $i \in \mathcal{I}$.*

Thus, Proposition 4 is the direct counterpart of Proposition 3, substituting unit cost rates that are increasing in the state-of-the-world, for capacity levels that are decreasing in the latter (just like Theorem 5 is the direct counterpart of Theorem 4). One might conjecture that Propositions 1 and 2 have similar counterparts in terms of state-of-the-world dependent capacity levels, but counterexamples

show otherwise. For example, the counterpart of Proposition 2 fails to be true in general, as is shown by Figure 7 in Appendix D, where a more volatile capacity vector (higher β) leads to higher rather than lower cost for a fixed Markov chain. This occurs everywhere where the curves cross. We also provide an intuitive explanation in Appendix D on why this crossing phenomenon may occur.

6.3 Numerical Results

We now present several numerical studies to illustrate and complement the above theoretical results. Table 2 and Figure 4 in Appendix D evaluate the impact of different patterns of Markov-modulated cost rates, as discussed in Section 6.2, while Table 3 and Figure 6 (both in Appendix D) assess the impact of different patterns of Markov-modulated capacity limits. All instances have deterministic supplies. The base values of the various parameters are $l_e = 2, l_r = 3$, the per-period demand is Normally distributed with mean 5 and standard deviation 2, the holding and backlogging cost rates are $h = 5, p = 95$, respectively, and there are no salvage opportunities.

In choosing the base values of the model parameters, we have been guided by the choices made in Janakiraman et al. (2015)'s numerical study. $c_{n,e}^i$ should be interpreted as the unit differential between the expedited and regular suppliers. In a model with stationary parameters and the long-run cost per period criterion, only this cost differential is relevant when focusing only on the controllable part of the cost, as the long-run average procurement cost is equal to $(c_r \mathbb{E}D)$ irrespective of what control policy is followed. We thus, normalized $c_r = 0$ and vary the value of c_e around 20 or 25. We set $h = 5$. As stated in Janakiraman et al. (2015), with a review period of a month and a 20% annual cost of capital, this corresponds to a regular cost c_r of \$300. (In Janakiraman et al. (2015), h is chosen at the level of $h = 20$, which under the given annual cost of capital would correspond with a c_r value of \$1200. The p value of 95 corresponds with a service level of $\frac{p}{p+h} = 95\%$ in a single source model.)

Table 2: Optimal long-run average cost for different Markov-modulated systems

Transition matrix $c_{n,e}^i$	P_1		P_2		P_3	
	cost	diff ¹	cost	diff ¹	cost	diff ¹
{20, (20), 20, (20), 20}	83.84		83.84		83.84	
{10, (15), 20, (25), 30}	81.38	-2.9%	75.90	-9.5%	73.01	-12.9%
{0, (10), 20, (30), 40}	76.17	-9.1%	66.27	-21.0%	61.78	-26.3%
{25, (25), 25, (25), 25}	93.84		93.84		93.84	
{15, (20), 25, (30), 35}	91.38	-2.6%	85.90	-8.5%	83.01	-11.5%
{5, (15), 25, (35), 45}	86.17	-8.2%	76.27	-18.7%	71.78	-23.5%

$$P_1 = \left\{ \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\}, \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\}, \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\} \right\}$$

$$P_2 = \left\{ \left\{ \frac{2}{3}, \frac{1}{3}, 0 \right\}, \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\}, \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\} \right\}$$

$$P_3 = \left\{ \left\{ \frac{3}{5}, \frac{2}{5}, 0, 0, 0 \right\}, \left\{ \frac{1}{5}, \frac{3}{5}, \frac{1}{5}, 0, 0 \right\}, \left\{ \frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{5}, 0 \right\}, \left\{ \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right\}, \left\{ \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right\} \right\}$$

$$C_e = C_r = 3$$

¹ Relative difference to the (bold) benchmark costs

With $C_e = C_r = 3$, Table 2 compares the optimal long-run average cost for systems with different Markov chains governing the state-of-the-world and different Markov-modulated unit costs of the *expedited* supplier. It is easily verified that the optimal long-run average cost is a constant, independent of the starting inventory position or state-of-the-world. There are three Markov chains with transition matrices P_1, P_2 and P_3 specified below the table, where P_1 and P_2 have three states and P_3 has five states. P_1 and P_2 have states $\{1, 3, 5\}$, while P_3 has states $\{1, 2, 3, 4, 5\}$. Under each Markov system, two sets of expedited costs are evaluated, one centered around $c_{n,e} = 20$ and the other around $c_{n,e} = 25$.

Comparing rows in the upper and lower half of Table 2 under each Markov chain, we find that the optimal long-run average cost is lower when c_e fluctuates more. For instance, under the three-state Markov chain P_1 with equal transition probabilities, compared with the stable price $c_{n,e}^i \in \{20, 20, 20\}$, the cost is 2.9% lower under the volatile price $c_{n,e}^i \in \{10, 20, 30\}$ and 9.1% lower under the even more volatile price $c_{n,e}^i \in \{0, 20, 40\}$, illustrating Proposition 1. The fact that the cost vector $\{0, 20, 40\}$ results in approximately 6% additional cost savings beyond those achieved under vector $\{10, 20, 30\}$ is an illustration of Proposition 2. The cost savings are even more significant under Markov chains P_2 and P_3 , as the probabilities of staying in good states are higher, which reflects Proposition 3. Similar patterns appear in the sets of expedited unit costs with average value $c_{n,e} = 25$.

Figure 4 in Appendix D depicts the value function $f_n^i(x)$ under four settings, selected from the 18 instances from Table 2, all relating to period $n = 10$. In each subfigure, it is clear that the cost-to-go is ranked by the initial state-of-the-world for any inventory position x , thus illustrating Theorem 4. The comparisons of subfigures (a) vs (b) and (c) vs (d) both show the value functions are lower in the “more volatile price world” ((b) and (d)) than in the corresponding “more stable price world” ((a) and (c)), thus illustrating Proposition 2.

It is, of interest, to investigate how the cost savings depend on the degree of dispersion in the cost rates, and the volatility of the world state Markov chain. The former is defined as

$$\frac{\frac{1}{2}\text{span}(c_{n,e}^1, \dots, c_{n,e}^I)}{\text{avg}(c_{n,e}^1, \dots, c_{n,e}^I)},$$

where $\text{span}(x) = x_{\max} - x_{\min}$ and $\text{avg}(x) = \frac{1}{I} \sum_{i=1}^I x_i$. This measure is a dimensionless index. To this end, we consider the parameterized cost vector $(20(1-\beta), 20, 20(1+\beta))$ for some $\beta \in [0, 1]$. Note that β equals the above dispersion measure. To characterize the volatility of a Markov chain A, relative to one (B) where the steady-state distribution is uniform in the state space, we use the total variation distance measure (where $\pi^{(A)}, \pi^{(B)}$ are the steady-state distributions)

$$\frac{1}{2} \sum_{i=1}^I |\pi_i^{(A)} - \pi_i^{(B)}|.$$

In Appendix D, we display 12 subfigures in Figure 5, varying one parameter at a time from the base case. Not surprisingly, we observe, throughout the parameter space, that the cost savings increase with

the total variation distance. Note that we design the transition matrix such that, the larger the total variation distance, the higher the steady-state probability of being in the good state. Also, throughout all the graphs, the cost savings increase in β , the degree of dispersion in the cost rates. Depending on the “total variation distance” and the β value, the cost savings can be extensive and in the 10–20% range, sometimes larger. For a given value of $p[h]$, the relative cost savings, are all the more impressive as $p[h]$ declines, presumably because the relative weight of the purchase costs increases.

In Appendix D, we also investigate the impact of various patterns by which *capacity limits* vary with the state-of-the-world, and under different state-of-the-world Markov chains. See there for the details and managerial insights.

7 Conclusions and Generalizations

We have proposed and analyzed a very general periodic review model for dual sourcing. The model addresses a broad spectrum of volatilities encountered in practice, partitioned into *internal* and *external* volatilities. These include cost structures, capacity levels and demand distributions which fluctuate in general ways in response to *external* environmental (economic, political, product life cycle, etc.) factors. We also include volatilities in the demand distributions and the supply mechanisms, where targeted orders are subject to random curtailments due to random yields or random capacities, among others. To our knowledge, this is the first model that addresses the choice between suppliers who are differentiated by the prices they charge, the reliability of their supply mechanisms as well as their lead times and capacities. The model allows for salvage opportunities and general capacity limits.

In this general model, it is important to distinguish between settings where all orders in a given period need to be placed simultaneously and those where the determination of the regular orders may await the resolution of any uncertainty surrounding the actual delivery by the expedited supplier. We have characterized the structure of an optimal strategy, under parallel and sequential ordering, giving special treatment to the random capacity model (4a) after covering the case with general random supply mechanisms that satisfy the SL(mp) property in Assumption 3. We have also derived conditions, for cost rates and capacity levels, under which the value functions are monotone in the initial state-of-the-world.

As a second set of contributions, we have compared and contrasted the impact of internal vs external volatilities. Numerical examples illustrate these comparison results and reject other reasonable conjectures. We show in particular that, volatilities arising from the fluctuating state-of-the-world, which leave *expected* prices or capacity levels unchanged, may be exploited to *reduce* system-wide costs, by reallocating orders along with adjustment of the aggregate orders. This explains the power of multi-sourcing, especially in volatile world environments.

As explained in the introduction, the biggest restriction in our work is that it assumes lead times

are consecutive. Under general lead time combinations, fully optimal policies necessarily depend on a vector of inventory measures, as opposed to single dimensional inventory measures, here. However, the structural results obtained under consecutive lead times, along with other practical considerations, motivate using the same classes of optimal policies (e.g., modified base-stock policies) even under general lead times. This reduces the challenge to searching for an optimal combination of policy *parameters* (e.g., base-stock levels) with simulation optimization, see e.g. Fu (2015).

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Appendix – Dual Sourcing under Internal and External Volatilities

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A Proofs

A.1 Proofs of Lemmas

Proof of Lemma 1. (a) We first show the value functions $f_n^i(\cdot)$ are convex and $|f_n^i(x)| = O(|x|^p)$ by induction. By assumption, the statement holds for $n = 0$. Suppose f_{n-1}^j is convex and $|f_{n-1}^j(x)| = O(|x|^p)$ for some n and all $j \in \mathcal{I}$. Since $\mathcal{U}_{n,e}^i$ and $\mathcal{U}_{n,r}^i$ are simple closed intervals that are independent of x , to show that f_n^i is convex for all $i \in \mathcal{I}$, it suffices, employing standard arguments, to show that $\mathbb{E}L_n^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e))$ is convex in (x, q_v, μ_e) and that $\mathbb{E}f_{n-1}^j(x + q_v + \hat{Y}_{n,e}^i(\mu_e) + \hat{Y}_{n,r}^i(\mu_r) - D_n^i)$ is convex in (x, q_v, μ_e, μ_r) . We first show the former.

Note that $\mathbb{E}L_n^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e))$ is continuous in (x, q_v, μ_e) as the composition of continuous functions, see Assumptions 2-3 and (2). Thus, to show convexity it suffices to show midpoint convexity. Take any $x_1, x_2 \in \mathbb{R}$, $q_v^1, q_v^2 \in [-C_{n,v}^i, 0]$ and $\mu_e^1, \mu_e^2 \in \mathcal{U}_{n,e}^i$. From Definition 1, we have

$$\frac{\hat{Y}_{n,e}^i(\mu_e^1) + \hat{Y}_{n,e}^i(\mu_e^2)}{2} \leq_{\text{concave}} \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right),$$

and hence

$$\frac{x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1) + x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2)}{2} \leq_{\text{concave}} \frac{x_1 + x_2}{2} + \frac{q_v^1 + q_v^2}{2} + \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right).$$

By the convexity of $L_n^i(\cdot)$ and the SL(mp) property of the supply mechanisms $\hat{Y}_{n,e}^i(\cdot)$ (see Definition 1), we have

$$\mathbb{E}L_n^i\left(\frac{x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1) + x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2)}{2}\right) \geq \mathbb{E}L_n^i\left(\frac{x_1 + x_2}{2} + \frac{q_v^1 + q_v^2}{2} + \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right)\right). \quad (23)$$

Moreover, for any realization of $\hat{Y}_{n,e}^i(\mu_e^1)$ and $\hat{Y}_{n,e}^i(\mu_e^2)$, since L_n^i is convex, we know

$$\frac{L_n^i(x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1)) + L_n^i(x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2))}{2} \geq L_n^i\left(\frac{x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1) + x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2)}{2}\right). \quad (24)$$

Since this inequality applies in every sample path, it applies in particular when we take expectations on both sides.

Combining Eqs. (23)–(24), we get

$$\frac{\mathbb{E}L_n^i(x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1)) + \mathbb{E}L_n^i(x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2))}{2} \geq \mathbb{E}L_n^i\left(\frac{x_1 + x_2}{2} + \frac{q_v^1 + q_v^2}{2} + \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right)\right). \quad (25)$$

Therefore, we have shown $\mathbb{E}L_n^i(x + q_v + \hat{Y}_{n,e}^i(\mu_e))$ is convex in x, q_v, μ_e .

It remains to be shown that

$$\mathbb{E}f_{n-1}^j(x + q_v + \hat{Y}_{n,e}^i(\mu_e) + \hat{Y}_{n,r}^i(\mu_r) - D_n^i) \text{ is convex in } (x, q_v, \mu_e, \mu_r). \quad (26)$$

We first establish the continuity of the above function, so that midpoint convexity is sufficient to establish convexity.

Let $y \equiv x + q_v + \hat{Y}_{n,e}^i(\mu_e) + \hat{Y}_{n,r}^i(\mu_r)$, and consider a sequence $\{x^{(m)}, q_v^{(m)}, \mu_e^{(m)}, \mu_r^{(m)}\}_{m=1}^\infty \rightarrow \{x^{(0)}, q_v^{(0)}, \mu_e^{(0)}, \mu_r^{(0)}\}$. Since $\{x^{(m)}\}_{m=1}^\infty$ is contained in a finite interval around $x^{(0)}$ and $q_v^{(m)} \in [-C_{n,v}^i, 0]$, $\mu_e^{(m)} \in \mathcal{U}_{n,e}^i$ and $\mu_r^{(m)} \in \text{TM}\mathcal{U}_{n,r}^i$, which are bounded intervals. Since y is a continuous function of the four variables (x, q_v, μ_e, μ_r) , an upper bound U exists such that $|y^{(m)}| \leq U$, for all m , where $y^{(m)} \equiv x^{(m)} + q_v^{(m)} + \hat{Y}_{n,e}^i(\mu_e^{(m)}) + \hat{Y}_{n,r}^i(\mu_r^{(m)})$. Then, since by the induction assumption, $|f_{n-1}^j(y)| \leq K|y|^p$ for some constant K , $|f_{n-1}^j(y^{(m)} - D_n^i)| \leq \sum_{l=0}^p \binom{p}{l} |D_n^i|^l (y^{(m)})^{p-l} \leq K \sum_{l=0}^p \binom{p}{l} |D_n^i|^l U^{p-l}$, which has a finite expectation over the distribution of D_n^i , by Assumption 2 (b). Thus, the condition for the dominated convergence theorem are satisfied and

$$\lim_{m \rightarrow \infty} \mathbb{E}f_{n-1}^j(x^{(m)} + q_v^{(m)} + \hat{Y}_{n,e}^i(\mu_e^{(m)}) + \hat{Y}_{n,r}^i(\mu_r^{(m)}) - D_n^i) = \mathbb{E}f_{n-1}^j(x^{(0)} + q_v^{(0)} + \hat{Y}_{n,e}^i(\mu_e^{(0)}) + \hat{Y}_{n,r}^i(\mu_r^{(0)}) - D_n^i),$$

establishing the continuity of the function in (26).

It thus suffices to show midpoint convexity, i.e.,

$$\begin{aligned} & \frac{\mathbb{E}f_{n-1}^j(x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1) + \hat{Y}_{n,r}^i(\mu_r^1) - D) + \mathbb{E}f_{n-1}^j(x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2) + \hat{Y}_{n,r}^i(\mu_r^2) - D)}{2} \\ & \geq \mathbb{E}f_{n-1}^j\left(\frac{x_1 + x_2}{2} + \frac{q_v^1 + q_v^2}{2} + \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right) + \hat{Y}_{n,r}^i\left(\frac{\mu_r^1 + \mu_r^2}{2}\right) - D\right) \end{aligned} \quad (27)$$

for $x_1, x_2 \in \mathbb{R}$, $q_v^1, q_v^2 \in [-C_{n,v}^i, 0]$, $\mu_e^1, \mu_e^2 \in \mathcal{U}_{n,e}^i$ and $\mu_r^1, \mu_r^2 \in \mathcal{U}_{n,r}^i$. Since $\{\hat{Y}_{n,e}^i(\mu_e), \mu_e \in \mathcal{U}_{n,e}^i\} \in \text{SL}(\text{mp})$ and $\{\hat{Y}_{n,r}^i(\mu_r), \mu_r \in \mathcal{U}_{n,r}^i\} \in \text{SL}(\text{mp})$, by Definition 1, we have for any $\mu_e^1, \mu_e^2 \in \mathcal{U}_{n,e}^i$,

$$\frac{\hat{Y}_{n,e}^i(\mu_e^1) + \hat{Y}_{n,e}^i(\mu_e^2)}{2} \leq_{\text{concave}} \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right),$$

and for any $\mu_r^1, \mu_r^2 \in \mathcal{U}_{n,r}^i$,

$$\frac{\hat{Y}_{n,r}^i(\mu_r^1) + \hat{Y}_{n,r}^i(\mu_r^2)}{2} \leq_{\text{concave}} \hat{Y}_{n,r}^i\left(\frac{\mu_r^1 + \mu_r^2}{2}\right).$$

Hence, take any $x_1, x_2 \in \mathbb{R}$, we have, by the independence assumptions that

$$\begin{aligned} & \frac{x_1 + q_v^1 + \hat{Y}_{n,e}^i(\mu_e^1) + \hat{Y}_{n,r}^i(\mu_r^1) - D}{2} + \frac{x_2 + q_v^2 + \hat{Y}_{n,e}^i(\mu_e^2) + \hat{Y}_{n,r}^i(\mu_r^2) - D}{2} \\ & \leq_{\text{concave}} \frac{x_1 + x_2}{2} + \frac{q_v^1 + q_v^2}{2} + \hat{Y}_{n,e}^i\left(\frac{\mu_e^1 + \mu_e^2}{2}\right) + \hat{Y}_{n,r}^i\left(\frac{\mu_r^1 + \mu_r^2}{2}\right) - D. \end{aligned}$$

Thus $f_n^i(x)$ is convex, and $|f_n^i(x)| = O(|x|^p)$ is established in analogy to (26), thus completing the induction proof.

The remainder of the proof of (27) is analogous to that of (25). The stochastic concavity in the above inequality follows from stochastic concavity being closed under convolutions, see Theorem 3.A.12 (d) in Shaked and Shanthikumar (2007).

The remaining proof of part (a) is analogous to that of Lemma 1 (a) and Lemma 1 (b) in Federgreen and Yang (2011).

(b) From the proof of part (a) and Assumption 2, the minimand in (6) is convex in (q_v, μ_e, μ_r) . The result hence follows. $\square$

Proof of Lemma 2. Let $F_n^i(x, q_v, \mu_e, \mu_r)$ denote the minimand in (6). Using Lemma 1 (b), consider the partial derivatives:

$$\begin{aligned} \frac{\partial F_n^i(x, q_v, \mu_e, \mu_r)}{\partial q_v} &= c_{n,v}^i + \mathbb{E}L_n^{i'}(x + q_v + \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)) \\ &+ \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x + q_v + \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i) + \phi_{n,r}^i(q_{n,r}^i(\mu_r), Z_{n,r}^i) - D_n^i) \end{aligned} \quad (28)$$

and

$$\begin{aligned} \frac{\partial F_n^i(x, q_v, \mu_e, \mu_r)}{\partial \mu_e} &= c_{n,e}^i + \mathbb{E}\left(L_n^{i'}(x + q_v + \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)) \frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(\mu_e)\right) \\ &+ \alpha \sum_{j=1}^I p_{ij} \mathbb{E}\left(f_{n-1}^j{}'(x + q_v + \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i) + \phi_{n,r}^i(q_{n,r}^i(\mu_r), Z_{n,r}^i) - D_n^i) \frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(\mu_e)\right). \end{aligned} \quad (29)$$

(The interchange of the expectation and derivative operators can be justified, since $|f_n^{j'}(x)| = O(|x|^p)$ and $\mathbb{E}[D_n^j]^p < \infty$.)

Let

$$G(Z_{n,e}^i) := L_n^{i'}(x + q_e + \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)) \quad \text{and} \quad H(Z_{n,e}^i) := \frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(\mu_e).$$

Supermodularity of the functions $\phi_{n,e}^i(x, Z)$ implies that the function $\frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e}$ is increasing in $Z_{n,e}^i$, i.e., $H(\cdot)$ is an increasing function, in view of (2). Similarly, $G(\cdot)$ is an increasing function, since $L_n^{i'}(\cdot)$ is increasing

while $\phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)$ is increasing in $Z_{n,e}^i$, by the assumption in our proposition. By a well-known inequality, due to Harris (1960), see e.g., Section 2.2 in Grimmett (1999), $\mathbb{E}[G(Z_{n,e}^i)H(Z_{n,e}^i)] \geq \mathbb{E}G(Z_{n,e}^i)\mathbb{E}H(Z_{n,e}^i) = \mathbb{E}G(Z_{n,e}^i)$, see (2). Since $c_{n,e}^i > c_{n,v}^i$, by Assumption 1, we have that each of the first two terms to the right of (28) is strictly dominated by the corresponding term to the right of (29). A similar argument shows that the third term to the right of (28) is dominated by the third term to the right of (29), since $f_{n-1}^j(\cdot)$ is increasing. We conclude that

$$\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} < \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e}.$$

If $q_v^* < 0$ is optimal, then since F_n^i is convex in (q_v, μ_e, μ_r) , by Lemma 1 (a) and Assumption 2, $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} = 0$, and by the above inequality, $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e} > 0$. This implies that $\mu_e^* = q_v^* \mu_r^* = 0$. $\square$

Proof of Lemma 3. Clearly, $x \in \mathcal{A}_{n,e}^i(x)$ [$y \in \mathcal{A}_{n,r}^i(y)$] since $\xi_{n,e}^i \in [-C_{n,e}^i, \infty)$ [$\xi_{n,r}^i \in [-C_{n,r}^i, \infty)$]. It suffices to show that the sets are convex. We prove convexity for the set $\mathcal{A}_{n,r}^i(y)$. The proof for the set $\mathcal{A}_{n,e}^i(x)$ is similar.

Fix any y . Take any two values v_1 and v_2 in $\mathcal{A}_{n,r}^i(y)$ and any $\lambda \in [0, 1]$. Since $v_1, v_2 \in \mathcal{A}_{n,r}^i(y)$, there exists $z_1, z_2 \geq y$ and $\xi_1, \xi_2 \in \mathcal{X}_{n,r}^i$ such that $v_1 = z_1 \wedge (y + C_{n,r}^i + \xi_1)$ and $v_2 = z_2 \wedge (y + C_{n,r}^i + \xi_2)$. To show $\lambda v_1 + (1 - \lambda)v_2 \in \mathcal{A}_{n,r}^i(y)$, we need to show there exists $z_3 \geq y$ and $\xi_3 \in \mathcal{X}_{n,r}^i$ such that $\lambda v_1 + (1 - \lambda)v_2 = z_3 \wedge (y + C_{n,r}^i + \xi_3)$.

Compute

$$\lambda v_1 + (1 - \lambda)v_2 = \lambda z_1 \wedge (\lambda y + \lambda C_{n,r}^i + \lambda \xi_1) + (1 - \lambda)z_2 \wedge ((1 - \lambda)y + (1 - \lambda)C_{n,r}^i + (1 - \lambda)\xi_2).$$

Since $\mathcal{X}_{n,r}^i$ is convex, we have $\lambda \xi_1 + (1 - \lambda)\xi_2 \in \mathcal{X}_{n,r}^i$. It is easy to verify that if $z_1 \leq y + C_{n,r}^i + \xi_1$ and $z_2 \leq y + C_{n,r}^i + \xi_2$, or if $z_1 \geq y + C_{n,r}^i + \xi_1$ and $z_2 \geq y + C_{n,r}^i + \xi_2$, then $\lambda v_1 + (1 - \lambda)v_2 = (\lambda z_1 + (1 - \lambda)z_2) \wedge (y + C_{n,r}^i + \lambda \xi_1 + (1 - \lambda)\xi_2) \in \mathcal{A}_{n,r}^i(y)$. Suppose $v_1 = z_1 \leq y + C_{n,r}^i + \xi_1$ and $v_2 = y + C_{n,r}^i + \xi_2 \leq z_2$ (the case $z_1 \geq y + C_{n,r}^i + \xi_1$ and $z_2 \leq y + C_{n,r}^i + \xi_2$ follows similarly). Define $z_3 = \lambda z_1 + (1 - \lambda)(y + C_{n,r}^i + \xi_2) \geq y$. Then $\lambda v_1 + (1 - \lambda)v_2 = z_3 \wedge (y + C_{n,r}^i + \lambda \xi_1 + (1 - \lambda)\xi_2) \in \mathcal{A}_{n,r}^i(y)$. $\square$

A.2 Proofs of Propositions

Proof of Proposition 1. (a) Consider the stable price world's optimal policy π . Apply this policy to system B, i.e., the volatile price world. Under policy π , the order quantities do not depend on the state-of-the-world. It is obvious that π is a feasible policy in system B. It suffices to show that π incurs the same expected costs in system B, as in system A, under any given demand trajectory.

Fix a starting inventory position x_N and a demand trajectory; under policy π , the inventory position trajectory $\{x_n : n = N, \dots, 1\}$ in system B is the same as in system A. Therefore we incur the same total expected cost except, possibly, for the total expected ordering cost from the expedited supplier. Since the state-of-the-world Markov chain evolves independently from the demands, we can calculate the expected ordering cost rate from the expedited supplier in period n , which is bounded from above by c_n^* , by the theorem's assumption, and hence, identical to the unit cost rate in the stable price world. Therefore, the total expected (discounted) ordering cost from the expedited supplier is also identical in both worlds.

(b) This part is an immediate corollary of part (a). $\square$

Proof of Proposition 2. Let π^* denote the optimal policy under system A, and $\tilde{\pi}^*$ the policy in system B which in period n prescribes the same order decisions as π^* in state (x, i) , but is followed by the optimal policy for system B in subsequent periods. Since this is a feasible policy in system B,

$$\begin{aligned} f_n^i(x) - \tilde{f}_n^i(x) &\geq \mu_{n,e}^*(x, i)(c_{n,e}^i - \tilde{c}_{n,e}^i) \\ &+ \alpha \sum_{j=1}^I p_{ij} \mathbb{E}[f_{n-1}^j(x + \hat{Y}_{n,e}^i(\mu_{n,e}^*(x, i)) + q_{n,v}^*(x, i) + \hat{Y}_{n,r}^i(\mu_{n,r}^*(x, i)) - D_n^i) \\ &- \tilde{f}_{n-1}^j(x + \hat{Y}_{n,e}^i(\mu_{n,e}^*(x, i)) + q_{n,v}^*(x, i) + \hat{Y}_{n,r}^i(\mu_{n,r}^*(x, i)) - D_n^i)], \end{aligned} \quad (30)$$

where $\mu_{n,e}^*(x, i)$, $q_{n,v}^*(x, i)$, $\mu_{n,r}^*(x, i)$ denote the targeted order sizes and salvage quantity prescribed by policy π^* when (x, i) is the starting state in period n . Since the state $i_n = i \leq k$, the first term in (30) is non-negative. It suffices to show that the second term is non-negative as well.

Given a starting inventory position $x_n = x$ in period n , let $\{x_r : r = n-1, \dots, 1\}$ denote the inventory position process under policy π^* . We prove, by induction, that for all $r = 1, \dots, n$ and $i \in \mathcal{I}$:

$$\sum_{j=1}^I p_{ij} \mathbb{E}[f_{r-1}^j(x_{r-1}) - \tilde{f}_{r-1}^j(x_{r-1})] \geq 0. \quad (31)$$

Since $f_0^j(\cdot) = \tilde{f}_0^j(\cdot)$ for all $j \in \mathcal{I}$, (31) holds for $r = 1$. Assume it holds for some $r = 1, \dots, n-2$. Then

$$\begin{aligned} \sum_{j=1}^I p_{ij} \mathbb{E}[f_r^j(x_r) - \tilde{f}_r^j(x_r)] &\geq \sum_{j=1}^I p_{ij} \mu_e^*(x_r, j) [c_{n,e}^j - \tilde{c}_{n,e}^j] + \sum_{j=1}^I p_{ij} \sum_{l=1}^I p_{jl} \mathbb{E}[f_{r-1}^l(x_{r-1}) - \tilde{f}_{r-1}^l(x_{r-1})] \\ &\geq \sum_{j=1}^k p_{ij} \mu_e^*(x_r, j) [c_{n,e}^j - \tilde{c}_{n,e}^j] + \sum_{j=k+1}^I p_{ij} \mu_e^*(x_r, j) [c_{n,e}^j - \tilde{c}_{n,e}^j] \\ &\geq \mu_e^*(x_r, k+1) \left[\sum_{j=1}^k p_{ij} [c_{n,e}^j - \tilde{c}_{n,e}^j] + \sum_{j=k+1}^I p_{ij} [c_{n,e}^j - \tilde{c}_{n,e}^j] \right] \\ &= \mu_e^*(x_r, k+1) \sum_{j=1}^I p_{ij} [c_{n,e}^j - \tilde{c}_{n,e}^j] \geq 0. \end{aligned}$$

The first inequality follows by bounding $\tilde{f}_r^j(x_r)$, in system B, by the expected total (discounted) cost under the feasible policy which, in period r , adopts the order sizes prescribed by policy π^* , but follows system B's optimal policy thereafter. The second inequality follows from the induction assumption; and the third inequality from the monotonicity of $\{\mu_e^*(x_r, j)\}$ in j and (19). Finally, the last inequality results from (20). We have completed the induction proof for (31), and hence the proof for the theorem. $\square$

Proof of Proposition 3. By induction. The claim holds for $n = 0$. Assume it holds for some $n < N$. Fix a state-of-the-world $i \in \mathcal{I}$ and $x \in \mathbb{R}$. Let μ_r^*, μ_e^* and q_v^* denote the optimal targeted order sizes and salvage batch in system A in period $n+1$ where (i, x) is the state of the system. Then

$$\begin{aligned} f_{n+1}^i(x) &= c_{n+1,r}^i \mu_r^* + c_{n+1,e}^i \mu_e^* + c_{n+1,v}^i q_v^* + \mathbb{E} L_{n+1}^i(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^*) \\ &\quad + \alpha \mathbb{E}_{[i_n | i_{n+1}=i]} \mathbb{E}_{D_{n+1}^i, \hat{Y}_{n,e}^i(\mu_e^*), \hat{Y}_{n,r}^i(\mu_r^*)} f_n^{i_n}(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_{n+1}^i). \end{aligned}$$

By Theorem 4, $f_n^j(\cdot)$ is pointwise increasing in j , since (13) is satisfied and $\{i_n\}$ satisfy (21). This implies that $\mathbb{E}_{D_{n+1}^i, \hat{Y}_{n,e}^i(\mu_e^*), \hat{Y}_{n,r}^i(\mu_r^*)} f_n^j(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_{n+1}^i)$ is also increasing in j . Thus, since $(i_n | i_{n+1} = i) \geq_{st} (\tilde{i}_n | \tilde{i}_{n+1} = i)$, we get:

$$\begin{aligned} f_{n+1}^i(x) &\geq c_{n+1,r}^i \mu_r^* + c_{n+1,e}^i \mu_e^* + c_{n+1,v}^i q_v^* + \mathbb{E} L_{n+1}^i(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^*) \\ &\quad + \alpha \mathbb{E}_{[\tilde{i}_n | \tilde{i}_{n+1}=i]} \mathbb{E}_{D_{n+1}^i, \hat{Y}_{n,e}^i(\mu_e^*), \hat{Y}_{n,r}^i(\mu_r^*)} \tilde{f}_n^{\tilde{i}_n}(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_{n+1}^i) \\ &\geq c_{n+1,r}^i \mu_r^* + c_{n+1,e}^i \mu_e^* + c_{n+1,v}^i q_v^* + \mathbb{E} L_{n+1}^i(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^*) \\ &\quad + \alpha \mathbb{E}_{[\tilde{i}_n | \tilde{i}_{n+1}=i]} \mathbb{E}_{D_{n+1}^i, \hat{Y}_{n,e}^i(\mu_e^*), \hat{Y}_{n,r}^i(\mu_r^*)} \tilde{f}_n^{\tilde{i}_n}(x + \hat{Y}_{n,e}^i(\mu_e^*) + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_{n+1}^i) \\ &\geq \tilde{f}_{n+1}^i(x), \end{aligned}$$

where the second inequality follows from $f_n^i(\cdot) \geq \tilde{f}_n^i(\cdot)$ by the induction assumption, and the third inequality follows from the fact that the triplet $(\mu_r^*, \mu_e^*, q_v^*)$ is feasible but possibly suboptimal in system B. This completes the induction proof. $\square$

Proof of Proposition 4. Analogous to that of Proposition 3. $\square$

A.3 Proofs of Theorems

Proof of Theorem 1. We first establish that the three critical values $x_{n,0}^i, x_{n,e}^i$ and $x_{n,r}^i$ are well defined. Note that the left hand side of (7) is the derivative of the function $H(x) := c_{n,v}^i x + L_n^i(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x - D_n^i)$, which is convex by Assumption 2 and Lemma 1 (a) establishing the convexity of the functions $\{f_{n-1}^j(\cdot)\}$. Since the functions $H(\cdot)$ are convex, they are differentiable almost everywhere (except possibly for a countable set of values).

The interchange of the expectation and derivative operators is justified by a standard result, see e.g. Theorem 9.42 in [Rudin \(1976\)](#). This implies that $H'(\cdot)$, the function to the left of (7), is strictly increasing since $H(\cdot)$ is strictly convex. Moreover,

$$\lim_{x \downarrow -\infty} H'(x) \leq c_{n,v}^i + \lim_{x \downarrow -\infty} L_n^{i'}(x) \leq c_{n,e}^i + \lim_{x \downarrow -\infty} L_n^{i'}(x) < 0, \quad (32)$$

$$\lim_{x \uparrow \infty} H'(x) > c_{n,v}^i + \lim_{x \uparrow \infty} L_n^{i'}(x) > c_{n,v}^i + (-c_{n,v}^i)^+ \geq 0, \quad (33)$$

where the first inequalities in (32) and (33) follow from Lemma 1 (a) and the second inequality in (32) follows from Assumption 1 (i). The last inequality in (32) and the second inequality in (33) both follow from Assumption 2 (c). These properties ensure that function $H'(\cdot)$ has a *unique* value $-\infty < x_{n,0}^i < \infty$ such that $H'(x) < 0$ [$H'(x) > 0$] for $x < x_{n,0}^i$ [$x > x_{n,0}^i$]. We call $x_{n,0}^i$ the unique “root” of the function $H'(\cdot)$.

The proof that (8) has a unique finite root $-\infty < x_{n,e}^i < \infty$ is analogous. (Note the last inequality in (32)) Moreover, the function to the left of (7) is pointwise smaller than that to the left of (8) since, by Assumption 1, $c_{n,v}^i \leq c_{n,e}^i$. This implies that the unique root of the former is larger than that of the latter:

$$x_{n,0}^i \geq x_{n,e}^i.$$

Finally, by the above arguments, the function $\tilde{H}(x) = c_{n,r}^i x + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x - D_n^i)$ has a derivative function $\tilde{H}'(x) = c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x - D_n^i)$, which is increasing, and by Lemma 1 (a) positive for x sufficiently large. Thus, $x_{n,r}^i$ is either the unique root of $\tilde{H}(x) = 0$, or $\tilde{H}(x) > 0$ for all x , and $x_{n,r}^i = -\infty$.

- (a) We first show that for any $x > x_{n,0}^i$, it is optimal to place a salvage order $q_v^*(x) = (x_{n,0}^i - x) \vee (-C_{n,v}^i)$ and place no expedited/regular orders, i.e., $\mu_e^*(x) = \mu_r^*(x) = 0$. Substituting these values into (28) and (29), we get

$$\begin{aligned} & \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} \\ &= \begin{cases} c_{n,v}^i + L_n^{i'}(x_{n,0}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x_{n,0}^i - D_n^i), & \text{if } x \in [x_{n,0}^i, x_{n,0}^i + C_{n,v}^i], \\ c_{n,v}^i + L_n^{i'}(x - C_{n,v}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x - C_{n,v}^i - D_n^i), & \text{if } x > x_{n,0}^i + C_{n,v}^i, \end{cases} \end{aligned}$$

and

$$\begin{aligned} & \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e} \\ &= \begin{cases} c_{n,e}^i + L_n^{i'}(x_{n,0}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} \left(f_{n-1}^{j'}(x_{n,0}^i - D_n^i) \frac{\partial \phi_{n,e}^{i,(0,Z_{n,e}^i)}(q_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(0) \right), & \text{if } x \in [x_{n,0}^i, x_{n,0}^i + C_{n,v}^i], \\ c_{n,e}^i + L_n^{i'}(x - C_{n,v}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} \left(f_{n-1}^{j'}(x - C_{n,v}^i - D_n^i) \frac{\partial \phi_{n,e}^{i,(0,Z_{n,e}^i)}(q_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(0) \right), & \text{if } x > x_{n,0}^i + C_{n,v}^i. \end{cases} \end{aligned}$$

By the independence of D_n^i and Z_e , as well as (2), these expressions can be simplified to

$$\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e} = \begin{cases} c_{n,e}^i + L_n^{i'}(x_{n,0}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x_{n,0}^i - D_n^i), & \text{if } x \in [x_{n,0}^i, x_{n,0}^i + C_{n,v}^i], \\ c_{n,e}^i + L_n^{i'}(x - C_{n,v}^i) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x - C_{n,v}^i - D_n^i), & \text{if } x > x_{n,0}^i + C_{n,v}^i. \end{cases}$$

By similar arguments one verifies that

$$\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r} = \begin{cases} c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x_{n,0}^i - D_n^i), & \text{if } x \in [x_{n,0}^i, x_{n,0}^i + C_{n,v}^i], \\ c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^{j'}(x - C_{n,v}^i - D_n^i), & \text{if } x > x_{n,0}^i + C_{n,v}^i. \end{cases}$$

Since $x > x_{n,0}^i \geq \max\{x_{n,e}^i, x_{n,r}^i\}$, it follows from the definition of these thresholds that

$$\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e}, \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r} \geq 0,$$

while

$$\begin{aligned} & \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} > 0, \quad \text{if } x > x_{n,0}^i + C_{n,v}^i, \\ & \frac{\partial^+ F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} \geq 0, \quad \frac{\partial^- F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} \leq 0, \quad \text{if } x \leq x_{n,0}^i + C_{n,v}^i. \end{aligned}$$

This implies that the triplet $(q_v^*, \mu_e^*, \mu_r^*)$ satisfies the KKT conditions, and is, therefore, optimal in the convex program.

For $x \in [\max\{x_{n,r}^i, x_{n,e}^i\}, x_{n,0}^i]$, it is optimal not to take any action, i.e., the triplet $(q_v^* = 0, \mu_e^* = 0, \mu_r^* = 0)$ is optimal in (6). This follows immediately from the definitions of the threshold values so that

$$\begin{aligned}\frac{\partial F_n^i(x, 0, 0, 0)}{\partial q_v} &= c_{n,v}^i + L_n^{i'}(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x - D_n^i) \leq 0, \\ \frac{\partial F_n^i(x, 0, 0, 0)}{\partial \mu_e} &= c_{n,e}^i + L_n^{i'}(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x - D_n^i) \geq 0, \\ \frac{\partial F_n^i(x, 0, 0, 0)}{\partial \mu_r} &= c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x - D_n^i) \geq 0.\end{aligned}$$

Thus the triplet $(q_v^* = 0, \mu_e^* = 0, \mu_r^* = 0)$ again satisfies the KKT conditions.

The final case has $x < \max\{x_{n,e}^i, x_{n,r}^i\}$. We show that the optimal policy is to place no salvage orders and place some expedited/regular orders, i.e., $q_v^* = 0$ and $\mu_e^* + \mu_r^* > 0$. From Lemma 2, we know that $q_v^* \mu_e^* = 0$. Suppose $q_v^* < 0$. Then $\mu_e^* = 0$, and hence $\mu_r^* = 0$ cannot be optimal, since if it is, then $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v}$ reduces to $c_{n,v}^i + \mathbb{E} L_n^{i'}(x + q_v^*) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x + q_v^* - D_n^i)$ which is negative since $x < x_{n,0}^i$. This contradicts optimality. Therefore (under the assumption that $q_v^* < 0$) it must be that $\mu_e^* = 0$ and $\mu_r^* > 0$. We prove that

$$c_{n,v}^i + L_n^{i'}(x_{n,0}^i) \leq c_{n,r}^i. \quad (34)$$

When $x_{n,0}^i = x_{n,r}^i$, (34) holds as an equality by their definitions in (8) and (9). It remains to prove (34) when $x_{n,0}^i > x_{n,r}^i$. But in that case, let $\{y_t\}_{t=1}^\infty \uparrow x_{n,0}^i$ with $y_t > x_{n,r}^i$. Then

$$\frac{\partial F_n^i(y_t, 0, 0, 0)}{\partial q_v} = c_{n,v}^i + L_n^{i'}(y_t) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(y_t - D_n^i) < 0,$$

so that

$$\lim_{t \rightarrow \infty} \frac{\partial F_n^i(y_t, 0, 0, 0)}{\partial q_v} = c_{n,v}^i + L_n^{i'}(x_{n,0}^i) + \alpha \lim_{t \rightarrow \infty} \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(y_t - D_n^i) \leq 0.$$

At the same time,

$$\frac{\partial F_n^i(y_t, 0, 0, 0)}{\partial \mu_r} = c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(y_t - D_n^i) > 0,$$

so that

$$\lim_{t \rightarrow \infty} \frac{\partial F_n^i(y_t, 0, 0, 0)}{\partial \mu_r} = c_{n,r}^i + \alpha \lim_{t \rightarrow \infty} \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(y_t - D_n^i) \geq 0.$$

which imply (34). Hence $c_{n,v}^i + L_n^{i'}(x + q_v^*) \leq c_{n,v}^i + L_n^{i'}(x) < c_{n,r}^i$ for any $x < x_{n,0}^i$, since $L_n^{i'}(\cdot)$ is strictly increasing by Assumption 2. Thus by (2), since $x + q_v^* < x_{n,0}^i$, $c_{n,v}^i + L_n^{i'}(x + q_v^*) < c_{n,v}^i + L_n^{i'}(x_{n,0}^i) = c_{n,r}^i$ and

$$\begin{aligned}\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} &= c_{n,v}^i + L_n^{i'}(x + q_v^*) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_n^i) \\ &< c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j{}'(x + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_n^i) \cdot \mathbb{E} \frac{\partial \phi_{n,r}^i(q_r(\mu_r^*), Z_{n,r}^i)}{\partial q_r} q_r'(\mu_r^*) \\ &= c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} \left[f_{n-1}^j{}'(x + q_v^* + \hat{Y}_{n,r}^i(\mu_r^*) - D_n^i) \frac{\partial \phi_{n,r}^i(q_r(\mu_r^*), Z_{n,r}^i)}{\partial q_r} \right] q_r'(\mu_r^*) \\ &= \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r} = 0,\end{aligned}$$

where the second equality follows from the independence of Z_r and D_n^i , and the last equality follows from the optimality of $\mu_r^* > 0$. However, $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} < 0$, which contradicts the optimality of $q_v^* < 0$. Therefore, $q_v^* = 0$. It only remains to be shown that $\mu_e^* + \mu_r^* > 0$. Suppose this is not true, i.e., $\mu_e^* = \mu_r^* = 0$.

Then since $x < x_{n,e}^i \vee x_{n,r}^i$, at least one of $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_e}$ and $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r}$ is negative by the definition of $x_{n,0}^i$ and $x_{n,r}^i$. This contradicts $\mu_e^* = \mu_r^* = 0$.

- (b) If $x > x_{n,r}^i$, in this case, one should salvage, i.e., $q_v^* < 0$. Suppose $q_v^* = 0$, then

$$\begin{aligned} 0 &\geq \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} = c_{n,v}^i + \mathbb{E}L_n^{i'}(x + \hat{Y}_{n,e}^i(\mu_e^*)) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x + \hat{Y}_{n,e}^i(\mu_e^*) + \hat{Y}_{n,r}^i(\mu_r^*) - D_n^i) \\ &\geq c_{n,v}^i + \mathbb{E}L_n^{i'}(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x - D_n^i) > 0, \end{aligned}$$

a contradiction. (The first inequality follows from the KKT conditions, the second inequality follows from $\{\hat{Y}_{n,e}^i(\mu)\}$ and $\{\hat{Y}_{n,r}^i(\mu)\}$ being stochastically increasing in μ (see (3)), and $L_n^{i'}(\cdot)$ and all $f_{n-1}^j{}'(\cdot)$ being increasing functions, the former strictly increasing. The last inequality follows from $x > x_{n,0}^i$ and $L_n^{i'}(\cdot)$ being strictly increasing.) Therefore, it must be that $q_v^* < 0$ and hence $\mu_e^* = 0$, by Lemma 2.

We now show that for any $x \in (x_{n,0}^i, x_{n,r}^i)$, it is optimal to have $q_v^* < 0$ and $\mu_r^* > 0$ at the same time. Suppose $(q_v^*, \mu_e^*, \mu_r^*)$ is such that $q_v^* = 0$. Recall that

$$\begin{aligned} 0 &\geq \frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v} = c_{n,v}^i + \mathbb{E}L_n^{i'}(x + \hat{Y}_{n,e}^i(\mu_e^*)) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x + \hat{Y}_{n,e}^i(\mu_e^*) + \hat{Y}_{n,r}^i(\mu_r^*) - D_n^i) \\ &\geq c_{n,v}^i + \mathbb{E}L_n^{i'}(x) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x - D_n^i) > 0, \end{aligned}$$

a contradiction. (The first inequality follows from the KKT conditions, the second inequality follows from $\{\hat{Y}_{n,e}^i(\mu)\}$ and $\{\hat{Y}_{n,r}^i(\mu)\}$ being stochastically increasing in μ (see (3)), and $L_n^{i'}(\cdot)$ and all $f_{n-1}^j{}'(\cdot)$ being increasing functions, the former strictly increasing. The last inequality follows from $x > x_{n,0}^i$ and $L_n^{i'}(\cdot)$ being strictly increasing. Therefore, it must be that $q_v^* < 0$ and hence $\mu_e^* = 0$, by Lemma 2.

If $\mu_r^* = 0$,

$$\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r} = c_{n,r}^i + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x + q_v^* - D_n^i) < 0,$$

since $x + q_v^* < x_{n,r}^i$ and since $x_{n,r}^i$ is the unique ‘‘root’’ of the increasing function to the left of (9). But $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r} < 0$ violates the optimality conditions, thus showing that $\mu_r^* > 0$, all while $q_v^* < 0$.

In the last case, when $x < x_{n,0}^i$, suppose $q_v^* < 0$. Then $\mu_e^* = 0$ by Lemma 2, and hence $\mu_r^* = 0$ cannot be optimal, since if it is, then $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial q_v}$ reduces to $c_{n,v}^i + \mathbb{E}L_n^{i'}(x + q_v^*) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j{}'(x + q_v^* - D_n^i)$ which is negative since $x < x_{n,0}^i$. This contradicts optimality. Therefore, if $q_v^* < 0$, it must be that $\mu_e^* = 0$ and $\mu_r^* > 0$. On the other hand, suppose $q_v^* = 0$. Then it must be that $\mu_e^* + \mu_r^* > 0$. Suppose this is not true, i.e., $\mu_e^* = \mu_r^* = 0$. Then since $x < x_{n,r}^i$, hence $\frac{\partial F_n^i(x, q_v^*, \mu_e^*, \mu_r^*)}{\partial \mu_r}$ is negative by the definition of $x_{n,r}^i$. This contradicts $\mu_e^* = \mu_r^* = 0$. $\square$

Proof of Theorem 2. (a) Analogous to the proof Lemma 1.

- (b) We first prove that $\mu_r^*(\cdot)$ is a decreasing function. For an arbitrary inventory position reference value U , let $y' \equiv U - y$. Then

$$f_{n,r}^i(y) = - \max_{\mu_r \in \mathcal{U}_{n,r}^i} \left\{ -c_{n,r}^i \mu_r - \alpha \sum_{j=1}^I p_{ij} \mathbb{E}f_{n-1}^j \left(U - y' + \hat{Y}_{n,r}^i(\mu_r) - D_n^i \right) \right\}. \quad (35)$$

In view of part (a), the functions $\{-f_{n-1}^j(\cdot)\}$ are concave. We will show that this implies that the maximand in (35) is supermodular in (y', μ_r) , so that, by Theorem 2.3 in Vives (1999), μ_r^* is increasing in y' , and hence decreasing in y . It suffices to show the supermodularity of one of the terms $-\mathbb{E}f_{n-1}^j \left(U - y' + \hat{Y}_{n,r}^i(\mu_r) - D_n^i \right)$. Note that its partial derivative with respect to y' is given by

$$\mathbb{E}f_{n-1}^j{}' \left(U - y' + \hat{Y}_{n,r}^i(\mu_r) - D_n^i \right) = \mathbb{E}f_{n-1}^j{}' \left(U - y' + \phi_{n,r}^i(q_{n,r}^i(\mu_r), Z_{n,r}^i) - D_n^i \right),$$

which is an increasing function of μ_r since $f_{n-1}^j{}'(\cdot)$ is increasing by part (a), $\phi_{n,r}^i(\cdot, Z_{n,r}^i)$ is increasing in its first argument and $q_{n,r}^i(\cdot)$ is increasing.

To prove the monotonicity properties of the first stage, rewrite (11) as

$$f_n^i(x) = \min_{x - C_{n,v}^i \leq \underline{x} \leq x, \mu_e \in \mathcal{U}_{n,e}^i} \left\{ c_{n,e}^i \mu_e + c_{n,v}^i \underline{x} - c_{n,v}^i x + \mathbb{E} L_n^i(\underline{x} + \hat{Y}_{n,e}^i(\mu_e)) + \mathbb{E} f_{n,r}^i(\underline{x} + \hat{Y}_{n,e}^i(\mu_e)) \right\}. \quad (36)$$

Applying a change of both the state variable x and the action variable $\underline{x}$ to $x' = U - x$ and $\bar{x} = U - \underline{x}$, respectively, (36) is rewritten as

$$\begin{aligned} f_n^i(x) &= \min_{x' \leq \bar{x} \leq x' + C_{n,v}^i, \mu_e \in \mathcal{U}_{n,e}^i} \left\{ c_{n,e}^i \mu_e + c_{n,v}^i x' - c_{n,v}^i \bar{x} + \mathbb{E} L_n^i(U - \bar{x} + \hat{Y}_{n,e}^i(\mu_e)) + \mathbb{E} f_{n,r}^i(U - \bar{x} + \hat{Y}_{n,e}^i(\mu_e)) \right\} \\ &= - \max_{x' \leq \bar{x} \leq x' + C_{n,v}^i, \mu_e \in \mathcal{U}_{n,e}^i} \left\{ -c_{n,e}^i \mu_e - c_{n,v}^i x' + c_{n,v}^i \bar{x} - \mathbb{E} L_n^i(U - \bar{x} + \hat{Y}_{n,e}^i(\mu_e)) - \mathbb{E} f_{n,r}^i(U - \bar{x} + \hat{Y}_{n,e}^i(\mu_e)) \right\}. \end{aligned} \quad (37)$$

By the same argument, used above, we show that the maximand in (37) is supermodular in $(\bar{x}, \mu_e)$. Since the action sets $\{(\bar{x}, \mu_e) : x' \leq \bar{x} \leq x' + C_{n,v}^i, \mu_e \in \mathcal{U}_{n,e}^i\}$ are increasing sets in x' , it follows from Theorem 2.3 in Vives (1999), combined with Remark 1, ibid, that $(\bar{x}^*, \mu_e^*)$ is increasing in x' , hence decreasing in x . In particular, $\mu_e^*(x)$ is decreasing, while $U - \bar{x}^*$ is decreasing in x , so that $\underline{x}^*$ is increasing in x .

- (c) We prove the optimality of the individual threshold policy in the first stage problem. The proof of the single threshold policy in the second stage problem is analogous. By an analogous proof to that of Proposition 2, we have

$$\mu_e^*(x) q_v^*(x) = 0 \quad (38)$$

for all x . Consider the minimand $F_n^i(x, q_v, \mu_e)$ of (11), and note that

$$\begin{aligned} \frac{\partial F_n^i(x, q_v, \mu_e)}{\partial \mu_e} &= c_{n,e}^i + \mathbb{E} L_n^{i'}(x + q_v + \hat{Y}_{n,e}^i(\mu_e)) \frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(\mu_e) \\ &\quad + \mathbb{E} f_{n,r}^{i'}(x + q_v + \hat{Y}_{n,e}^i(\mu_e)) \frac{\partial \phi_{n,e}^i(q_{n,e}^i(\mu_e), Z_{n,e}^i)}{\partial q_e} q_{n,e}^{i'}(\mu_e). \end{aligned}$$

In particular, with $\mu_e = 0$ we obtain, employing (2) and the fact that $\hat{Y}_{n,e}^i(0) = 0$ a.s.:

$$\frac{\partial F_n^i(x, q_v, 0)}{\partial \mu_e} = c_{n,e}^i + L_n^{i'}(x + q_v) + f_{n,r}^{i'}(x + q_v).$$

By Assumption 2 and part (a), this is a strictly increasing function of x and q_v . Moreover, again by Assumption 2 and part (a), we have

$$\lim_{x \downarrow -\infty} \frac{\partial F_n^i(x, 0, 0)}{\partial \mu_e} < c_{n,e}^i - c_{n,e}^i - \bar{c}_{n-1,r}/\alpha < 0, \quad \text{while} \quad \lim_{x \uparrow \infty} \frac{\partial F_n^i(x, 0, 0)}{\partial \mu_e} > c_{n,e}^i > 0.$$

This implies the existence of a finite threshold $x_1^* := \inf\{x : \frac{\partial F_n^i(x, 0, 0)}{\partial \mu_e} \geq 0\}$. For $x < x_1^*$, $\frac{\partial F_n^i(x, q_v^*, 0)}{\partial \mu_e} \leq \frac{\partial F_n^i(x, 0, 0)}{\partial \mu_e} < 0$ for any $q_v^* \leq 0$, so that the pair $(q_v^*, \mu_e^* = 0)$ fails to be optimal. This implies that $\mu_e^* > 0$. By Eq. (38), this implies that for $x < x_1^*$, $q_v^* = 0$. Similarly, for any μ_e ,

$$\begin{aligned} \frac{\partial F_n^i(x, 0, \mu_e)}{\partial q_v} &= c_{n,v}^i + \mathbb{E} L_n^{i'}(x + \hat{Y}_{n,e}^i(\mu_e)) + \mathbb{E} f_{n,r}^{i'}(x + \hat{Y}_{n,e}^i(\mu_e)) \\ &\geq c_{n,v}^i + \mathbb{E} L_n^{i'}(x) + \mathbb{E} f_{n,r}^{i'}(x) = \frac{\partial F_n^i(x, 0, 0)}{\partial q_v}, \end{aligned}$$

where the inequality follows from $\{\hat{Y}_{n,e}^i(\mu)\}$ being stochastically increasing in μ (see (3)), and $L_n^{i'}(\cdot)$ and $f_{n,r}^{i'}(\cdot)$ being increasing functions. By the above argument, $\frac{\partial F_n^i(x, 0, 0)}{\partial q_v}$ is, again, a strictly increasing function in x with $\lim_{x \downarrow -\infty} \frac{\partial F_n^i(x, 0, 0)}{\partial q_v} < c_{n,v}^i - c_{n,e}^i - \bar{c}_{n-1,r}/\alpha < 0$. Similarly, $\lim_{x \uparrow \infty} \frac{\partial F_n^i(x, 0, 0)}{\partial q_v} > c_{n,v}^i + (-c_{n,v}^i)^+ \geq 0$. This implies the existence of a finite threshold value $x_2^* := \inf\{x : \frac{\partial F_n^i(x, 0, 0)}{\partial q_v} \geq 0\}$. Moreover, since $c_{n,v}^i < c_{n,e}^i$, $\frac{\partial F_n^i(x, 0, 0)}{\partial q_v} < \frac{\partial F_n^i(x, 0, 0)}{\partial \mu_e}$ for all x , so that $x_2^* > x_1^*$. For $x > x_2^* > x_1^*$, $\frac{\partial F_n^i(x, 0, \mu_e^*)}{\partial q_v} \geq \frac{\partial F_n^i(x, 0, 0)}{\partial q_v} > 0$, so that the pair $(q_v^* = 0, \mu_e^*)$ cannot be optimal, hence $q_v^* < 0$. By Eq. (38), we have $\mu_e^* = 0$. Finally, for $x_1^* \leq x \leq x_2^*$, $q_v^* = \mu_e^* = 0$ since $(q_v^* = 0, \mu_e^* = 0)$ satisfies the KKT conditions. $\square$

Proof of Theorem 3. Fix $n = N, \dots, 1$ and state-of-the-world $i \in \mathcal{I}$.

We first show that the optimal regular order size in period n is $(z^* - y)^+$ where z^* is the minimizer of function $c_{n,r}^i z + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(z - D_n^i)$. Consider the optimization problem

$$\begin{aligned} f_{n,r}^i(y) &= \min_{z \geq y} \mathbb{E} \left[c_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) - y) + \alpha \sum_{j=1}^I p_{ij} f_{n-1}^j(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) - D_n^i) \right] \\ &= \min_{z \geq y} \mathbb{E} \left[c_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) + \alpha \sum_{j=1}^I p_{ij} f_{n-1}^j(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) - D_n^i) \right] - c_{n,r}^i y. \end{aligned}$$

Note that, without loss of optimality, the minimization may be confined to the interval $[y, \bar{z}(y)] = \mathcal{A}_{n,r}^i(y)$, see Lemma 3.

Recall the set $\mathcal{A}_{n,r}^i(y)$ is convex by Lemma 3. Define the function

$$F_{n,r}^i(z) := c_{n,r}^i z + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(z - D_n^i),$$

which is strictly convex by Theorem 2 (with strict convexity following from that of the $L_n^i(\cdot)$ functions by Assumption 2). Let z^* be its unique global minimizer. Let $z_{n,r}^i(y)$ be the unique minimizer of the strictly convex function $F_{n,r}^i(\cdot)$ on the convex set $\mathcal{A}_{n,r}^i(y)$. Next we show that

$$f_{n,r}^i(y) = \min_{z \geq y} \mathbb{E} \left[F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \right] - c_{n,r}^i y = \mathbb{E} \left[F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \right] - c_{n,r}^i y. \quad (39)$$

Fix any y and consider any $z \geq y$. It suffices to show $F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \geq F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i))$ for any $\xi_{n,r}^i \in \mathcal{X}_{n,r}^i$. Since $F_{n,r}^i(\cdot)$ is convex and $z_{n,r}^i(y)$ is its minimizer on the convex one-dimensional set $\mathcal{A}_{n,r}^i(y)$, the function $F_{n,r}^i(\cdot)$ is decreasing on $(y, z_{n,r}^i(y))$ and increasing on $(z_{n,r}^i(y), \bar{z}(y))$. Take any $\xi_{n,r}^i \in \mathcal{X}_{n,r}^i$. If $z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) \geq z_{n,r}^i(y)$, then $y + C_{n,r}^i + \xi_{n,r}^i \geq z_{n,r}^i(y)$, and hence $z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) \geq z_{n,r}^i(y) = z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i)$. Since $F_{n,r}^i(\cdot)$ is increasing on $(z_{n,r}^i(y), \bar{z}(y))$, this implies that $F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \geq F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i))$. On the other hand, if $z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) < z_{n,r}^i(y)$, since $z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) \leq y + C_{n,r}^i + \xi_{n,r}^i$, so we have $z \wedge (y + C_{n,r}^i + \xi_{n,r}^i) \leq z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i) \leq z_{n,r}^i(y)$. Since $F_{n,r}^i(\cdot)$ decreases on $(y, z_{n,r}^i(y))$, this implies that $F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \geq F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i))$. Therefore we have $F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \geq F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i))$ for any $z \geq y$ and $\xi_{n,r}^i \in \mathcal{X}_{n,r}^i$, and hence $z_{n,r}^i(y)$ achieves the minimum in (39). Therefore, for all $z \geq y$, $\mathbb{E} F_{n,r}^i(z \wedge (y + C_{n,r}^i + \xi_{n,r}^i)) \geq \mathbb{E} F_{n,r}^i(z_{n,r}^i(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i))$, proving (39).

Next we argue that $z_{n,r}^i(y) = \max\{z^*, y\}$. First, observe that the value z^* does not depend on y . If $y \geq z^*$, then, since the minimand in (39) is an increasing function on $[z^*, \infty)$, $z_{n,r}^i(y) = y$, i.e., the optimal regular order size is 0. On the other hand, if $y < z^*$, and if furthermore $\bar{z}(y) \leq z^*$, then $z_{n,r}^i(y) = \bar{z}(y)$. This means the optimal regular order size is $\bar{z}(y) - y$. Note that $\bar{z}(y) \geq y + C_{n,r}^i + \xi_{n,r}^i$, hence $\bar{z}(y) \wedge (y + C_{n,r}^i + \xi_{n,r}^i) = z^* \wedge (y + C_{n,r}^i + \xi_{n,r}^i)$ for any $\xi_{n,r}^i \in \mathcal{X}_{n,r}^i$, the optimal regular order size is equivalently $z^* - y$. In the last case, if $z^* \in (y, \bar{z}(y))$, then $z^* \in \mathcal{A}_{n,r}^i(y)$ and hence $z_{n,r}^i(y) = z^*$, i.e., the optimal regular order size is $z^* - y$.

Now we provide the proof for the first-stage policies. Consider the threshold level x_1^* identified in Theorem 2 (c), and let $x \in (-\infty, x_1^*)$. On this half line, we can repeat the proof above for the second stage, since for $x < x_1^*$ it is optimal not to salvage. The proof establishes the existence of a threshold $y_{n,i}^*$ such that $q_e^* = (y_{n,i}^* - x)^+$. It remains to be shown that $y_{n,i}^* = x_1^*$: indeed, $y_{n,i}^* < x_1^*$ would imply that $q_e^* = 0$ for $y_{n,i}^* < x < x_1^*$, contradicting Theorem 2 (c). Similarly, $y_{n,i}^* > x_1^*$ would imply that $q_e^* > 0$ for $x_1^* < x < y_{n,i}^*$, contradicting Theorem 2 (c) again.

Now consider $x \in (x_2^*, \infty)$. Here, only salvage batches need to be considered by Theorem 2 (c). By argument similar to the second stage, we identify a threshold $y_{n,i}^{**}$ such that $q_v^* = (y_{n,i}^{**} - x)^-$. Moreover, $y_{n,i}^{**} = x_2^*$ since either $y_{n,i}^{**} < x_2^*$ or $y_{n,i}^{**} > x_2^*$ generates a contradiction with Theorem 2 (c). $\square$

Proof of Theorem 4. By induction. The statement holds, by assumption, for $n = 0$. Assume it holds for some period $n - 1 \geq 0$. Select a pair of states-of-the-world $1 \leq i < j \leq I$ and let $\mu_{n,e}^{*j}(x)$, $q_{n,v}^{*j}(x)$ and $\mu_{n,r}^{*j}(x)$ denote the optimal order and salvage quantities in period n , in state (x, j) . Since the random supply functions are independent of the state-of-the-world, these decisions are also feasible when i is the state-of-the-world in period n . Thus,

$$f_n^i(x) \leq c_{n,r}^i \mu_{n,r}^{*j} + c_{n,e}^i \mu_{n,e}^{*j} + c_{n,v}^i q_{n,v}^{*j} + \mathbb{E} L_n(x + \hat{Y}_{n,e}^i(\mu_{n,e}^{*j}) + q_{n,v}^{*j})$$

$$\begin{aligned}
& + \alpha \mathbb{E}_{[i_{n-1}|i_n=i]} \mathbb{E}_{D_n} f_{n-1}^{i_{n-1}}(x + \hat{Y}_{n,e}^i(\mu_{n,e}^{*j}) + q_{n,v}^{*j} + \hat{Y}_{n,r}^i(\mu_{n,r}^{*j}) - D_n) \\
& \leq c_{n,r}^j \mu_{n,r}^{*j} + c_{n,e}^j \mu_{n,e}^{*j} + c_{n,v}^j q_{n,v}^{*j} + \mathbb{E} L_n(x + \hat{Y}_{n,e}^j(\mu_{n,e}^{*j}) + q_{n,v}^{*j}) \\
& \quad + \alpha \mathbb{E}_{[i_{n-1}|i_n=i]} \mathbb{E}_{D_n} f_{n-1}^{i_{n-1}}(x + \hat{Y}_{n,e}^j(\mu_{n,e}^{*j}) + q_{n,v}^{*j} + \hat{Y}_{n,r}^j(\mu_{n,r}^{*j}) - D_n) \\
& \leq c_{n,r}^j \mu_{n,r}^{*j} + c_{n,e}^j \mu_{n,e}^{*j} + c_{n,v}^j q_{n,v}^{*j} + \mathbb{E} L_n(x + \hat{Y}_{n,e}^j(\mu_{n,e}^{*j}) + q_{n,v}^{*j}) \\
& \quad + \alpha \mathbb{E}_{[i_{n-1}|i_n=j]} \mathbb{E}_{D_n} f_{n-1}^{i_{n-1}}(x + \hat{Y}_{n,e}^j(\mu_{n,e}^{*j}) + q_{n,v}^{*j} + \hat{Y}_{n,r}^j(\mu_{n,r}^{*j}) - D_n) = f_n^j(x).
\end{aligned}$$

Note that the first inequality follows from the fact that $\{\mu_{n,r}^{*j}, \mu_{n,e}^{*j}, q_{n,v}^{*j}\}$ is feasible in state (x, i) . The second inequality follows from the unit cost rankings (13) and the independence of $\hat{Y}_{n,e}^j(\mu)$ and $\hat{Y}_{n,r}^j(\mu)$ on j . And the third inequality follows from the stochastic ordering of $[i_{n-1}|i_n]$ in (14), combined with the induction assumption guaranteeing that the value functions $f_{n-1}^j(\cdot)$ are increasing in j . $\square$

Proof of Theorem 5. By induction. By assumption, the result is true for period $n = 0$. Assume it holds for some period $n - 1$. Fix some inventory position x in period n . Consider any two state $l \leq j$. By the capacity rankings in (15), the optimal decisions $\{\mu_{n,r}^{*j}, \mu_{n,e}^{*j}, q_{n,v}^{*j}\}$ in state (x, j) remain feasible in (x, l) . The remainder of the proof is analogous to that of Theorem 4, where the second inequality follows from the stochastic ordering of $\hat{Y}_{n,r}^j(\mu)$ and $\hat{Y}_{n,e}^j(\mu)$ in j , together with the fact that $L_n(\cdot)$ and $f_{n-1}^{i_{n-1}}(\cdot)$ are convex from Proposition 1, and that convex ordering is preserved for independent random variables under convolution. $\square$

Proof of Theorem 6. For systems $s = A, B$, let

$$f_{n,s}^i(x) = \min_{-C_{n,v}^i \leq q_v \leq 0, \mu_e \in \mathcal{U}_{n,e}^i, \mu_r \in \mathcal{U}_{n,r}^i} F_{n,s}^i(x, q_v, \mu_e, \mu_r), \quad (40)$$

where

$$\begin{aligned}
F_{n,s}^i(x, q_v, \mu_e, \mu_r) &= c_{n,v}^i q_v + c_{n,e}^i \mu_e + c_{n,r}^i \mu_r + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^{i,s}(q_e)) \\
& \quad + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1,s}^j(x + q_v + \hat{Y}_{n,e}^{i,s}(\mu_e) + \hat{Y}_{n,r}^{i,s}(\mu_r) - D_n^i).
\end{aligned}$$

The proof is by induction with respect to n . By assumption, $f_{0,B}^i(x) \geq f_{0,A}^i(x)$. Assume that, for some n , $f_{n-1,B}^i(x) \geq f_{n-1,A}^i(x)$ for all i and x . Since by Lemma 1 and Assumption 2, all $f_{n-1,B}^i(\cdot)$ and $L_n^i(\cdot)$ functions are convex, we have by (17)

$$\begin{aligned}
F_{n,B}^i(x, q_v, \mu_e, \mu_r) &= c_{n,v}^i q_v + c_{n,e}^i \mu_e + c_{n,r}^i \mu_r + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^{i,B}(q_e)) \\
& \quad + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1,B}^j(x + q_v + \hat{Y}_{n,e}^{i,B}(\mu_e) + \hat{Y}_{n,r}^{i,B}(\mu_r) - D_n^i) \\
& \geq c_{n,v}^i q_v + c_{n,e}^i \mu_e + c_{n,r}^i \mu_r + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^{i,A}(q_e)) \\
& \quad + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1,B}^j(x + q_v + \hat{Y}_{n,e}^{i,A}(\mu_e) + \hat{Y}_{n,r}^{i,A}(\mu_r) - D_n^i) \\
& \geq c_{n,v}^i q_v + c_{n,e}^i \mu_e + c_{n,r}^i \mu_r + \mathbb{E} L_n^i(x + q_v + \hat{Y}_{n,e}^{i,A}(q_e)) \\
& \quad + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1,A}^j(x + q_v + \hat{Y}_{n,e}^{i,A}(\mu_e) + \hat{Y}_{n,r}^{i,A}(\mu_r) - D_n^i) = F_{n,A}^i(x, q_v, \mu_e, \mu_r),
\end{aligned}$$

where the second inequality follows from the induction assumption. Since the minimands of (40) are pointwise ranked, it follows that $f_{n,B}^i(x) \geq f_{n,A}^i(x)$ for all i, x and n , completing the induction proof. $\square$

B Deterministic Supplies

In this appendix, we address the special case where supplies are *deterministic*, i.e., $Y_{n,e}^i(q) = q$ and $Y_{n,r}^i(q) = q$ almost surely. This is, of course, a special case of the random capacity structure under sequential ordering, characterized in the previous subsection (under deterministic supplies, orders may be placed sequentially without loss of generality). As shown in Theorem 3, modified base-stock policies are optimal to set the salvage quantity (if any), or the orders for the regular and expedited suppliers. In the case of deterministic supplies, we are able to determine the precise optimal allocation of the aggregate order between the two suppliers. Under deterministic supplies, we also characterize the structure of the optimal policies in the presence of fixed costs.

Under deterministic suppliers, the inventory position dynamics are given by

$$x_{n-1} = x_n + q_{n,e} + q_{n,v} + q_{n,r} - D_n^{i_n}.$$

Moreover, the *inventory level* IL_{n-l_e} at the *end* of period $n - l_e$ is given by

$$IL_{n-l_e} := x_n + q_{n,e} + q_{n,v} - D_{n,n-l_e}^{i_n}, \quad (41)$$

where $D_{n,m}^{i_n}$ is the aggregate demand in the time interval $[n, m]$ with $m \geq n$, and the state-of-the-world in period n is i_n .

Consider each period $n = N, \dots, 1$ and a state-of-the-world $i \in \mathcal{I}$. Under deterministic supplies, we can *project* the detailed procurement strategies onto ones dealing with the *aggregate* inventory position adjustment $Q = q_e + q_v + q_r$, considering the expected cost under the *optimal disaggregation* of Q among $0 \leq q_e \leq C_{n,e}^i$, $-C_{n,v}^i \leq q_v \leq 0$ and $0 \leq q_r \leq C_{n,r}^i$, such that $q_e + q_v + q_r = Q \in [-C_{n,v}^i, C_{n,e}^i + C_{n,r}^i]$. Note that

$$f_n^i(x) = \min_{-C_{n,v}^i \leq Q \leq C_{n,e}^i + C_{n,r}^i} \{G_n^i(x, Q) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(x + Q - D_n^i)\},$$

where

$$G_n^i(x, Q) = \min_{0 \leq q_e \leq C_{n,e}^i, -C_{n,v}^i \leq q_v \leq 0, 0 \leq q_r \leq C_{n,r}^i, q_e + q_v + q_r = Q} \{c_{n,e}^i q_e + c_{n,v}^i q_v + c_{n,r}^i q_r + L_n^i(x + q_e + q_v)\}. \quad (42)$$

The solution to (42) can be obtained virtually in closed form, providing much insight into how an “aggregate order” is optimally disaggregated among the three options, and how the procurement cost rates, capacity levels and the shape of the inventory/backlogging functions $L_n^i(\cdot)$ interact with each other in determining the optimal disaggregation.

Lemma 4 (Properties of the function $G_n^i(\cdot, \cdot)$). *Consider deterministic supplies. Fix a period n and a state-of-the-world i . Then*

- (a) *the function $G_n^i(x, Q)$ is jointly strictly convex in (x, Q) ;*
- (b) *the function $G_n^i(x, Q)$ is supermodular in (x, Q) .*

We now characterize the optimal solution to the disaggregation problem (42). In this case, $G_n^i(x, Q) = \min\{G_{n,1}^i(x, Q), G_{n,2}^i(x, Q)\}$, where

$$G_{n,1}^i(x, Q) = \min_{(Q - C_{n,r}^i)^+ \leq q_e \leq \min\{Q, C_{n,e}^i\}} \{c_{n,r}^i Q - \Delta_n^i q_e + L_n^i(x + q_e)\}, \quad (43)$$

$$G_{n,2}^i(x, Q) = \min_{\max\{Q - C_{n,r}^i, -C_{n,v}^i\} \leq q_v \leq \min\{Q, 0\}} \{c_{n,r}^i Q - \tilde{\Delta}_n^i q_v + L_n^i(x + q_v)\}, \quad (44)$$

with $\Delta_n^i := c_{n,r}^i - c_{n,e}^i$, the price difference between the suppliers, and $\tilde{\Delta}_n^i := c_{n,r}^i - c_{n,v}^i$, the difference between the regular supplier's price and the unit salvage revenue. The lower bounds in (43) and (44) follow from $Q - q_e = q_r \leq C_{n,r}^i$ and $Q - q_v = q_r \leq C_{n,r}^i$, respectively. The feasible regions are non-empty closed intervals since $-C_{n,v}^i \leq Q \leq C_{n,r}^i + C_{n,e}^i$, so that

$$C_{n,e}^i \geq (Q - C_{n,r}^i)^+ \quad \text{and} \quad Q \geq [Q - C_{n,r}^i]^+. \quad (45)$$

Proposition 5 (Optimal order disaggregation between two suppliers). *Consider deterministic supplies. Fix a period n and a state-of-the-world i . Let (q_r^*, q_e^*) denote the optimal solution of (43) that defines $G_{n,1}^i(x, Q)$. Without loss of generality, assume $Q \leq C_{n,r}^i + C_{n,e}^i$ (otherwise, no feasible solution exists). Let $x_1^*(Q) := (L_n^{i'})^{-1}(\Delta_n^i) - \min\{Q, C_{n,e}^i\}$ and $x_2^*(Q) := (L_n^{i'})^{-1}(\Delta_n^i) - [Q - C_{n,r}^i]^+ > x_1^*$.*

- (a) *If $x \leq x_1^*$:*

$$q_r^* = [Q - C_{n,e}^i]^+, \quad q_e^* = \min\{Q, C_{n,e}^i\}.$$

- (b) *If $x_1^* < x < x_2^*$:*

$$q_r^* = Q + x - (L_n^{i'})^{-1}(\Delta_n^i), \quad q_e^* = (L_n^{i'})^{-1}(\Delta_n^i) - x.$$

- (c) *If $x \geq x_2^*$:*

$$q_r^* = \min\{Q, C_{n,r}^i\}, \quad q_e^* = [Q - C_{n,r}^i]^+.$$

- (d) *$G_{n,1}^i(x, Q)$ is continuously differentiable with respect to x , and*

- for $x \leq x_1^*(Q)$, $\frac{\partial G_{n,1}^i(x,Q)}{\partial x} = L_n^{i'}(x + \min\{Q, C_{n,e}^i\})$;
- for $x_1^*(Q) < x < x_2^*(Q)$, $\frac{\partial G_{n,1}^i(x,Q)}{\partial x} = \Delta_n^i$;
- for $x \geq x_2^*(Q)$, $\frac{\partial G_{n,1}^i(x,Q)}{\partial x} = L_n^{i'}(x + [Q - C_{n,r}^i]^+)$.

Figure 3 shows that, for a given aggregate order, q_r^* is a piecewise linear increasing function of x , and q_e^* is a piecewise linear decreasing function, both with identical breakpoints at x_1^* and x_2^* . If the inventory position is sufficiently low, i.e., $x \leq x_1^*$, one allocates as much as possible of the total order Q to the expedited supplier. Similarly, if $x \geq x_2^*$, one allocates as much as possible to the regular supplier. In the middle region, $x_1^* < x < x_2^*$, neither one of the suppliers operates at capacity: q_r^* increases linearly (with slope +1) from its minimum level $[Q - C_{n,e}^i]^+$ to its maximum level $\min\{Q, C_{n,r}^i\}$, while q_e^* decreases linearly (with slope -1) from its maximum level $\min\{Q, C_{n,e}^i\}$ to its minimum level $[Q - C_{n,r}^i]^+$.

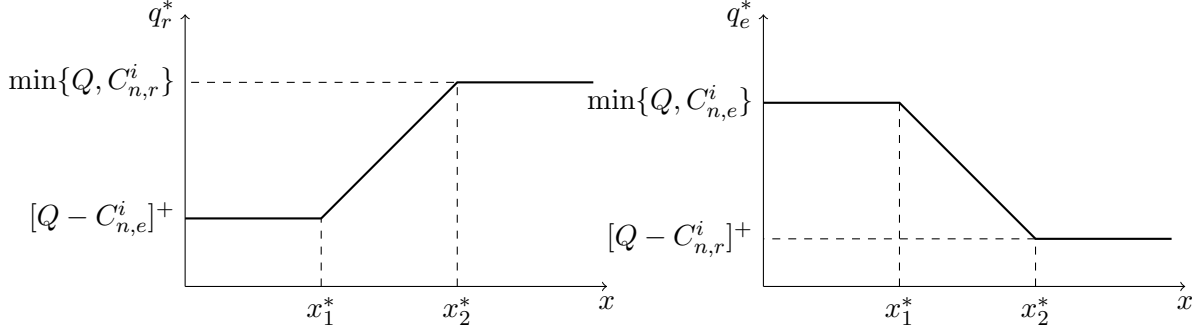


Figure 3: Dependence of q_r^* and q_e^* on the starting inventory position x

Proposition 5 also exhibits how various model parameters impact on the optimal allocation of aggregate orders between the two suppliers. For example, if in a given period n and state-of-the-world i , the unit price of the regular supplier increases by δ , this will cause both x_1^* and x_2^* to increase by an equal amount, $S = (L_n^{i'})^{-1}(\Delta_n^i + \delta) - (L_n^{i'})^{-1}(\Delta_n^i)$. As a consequence, maximum usage of her competitor (the expedited supplier) is extended over a larger region, while maximum usage of the regular supplier is curtailed to a smaller range of starting inventory positions. In the intermediate range, an amount S is shifted from the order with the regular supplier toward that of the expedited supplier. Corollary 3 characterizes the optimal solution of (44), which defines $G_{n,2}^i(x, Q)$.

Corollary 3 (Optimal order disaggregation between regular supplier and salvaging). *Fix a period n and a state-of-the-world i . Let (q_r^*, q_v^*) denote the optimal solution of (44) that defines $G_{n,2}^i(x, Q)$. Without loss of generality, assume $-C_{n,v}^i \leq Q \leq C_{n,r}^i$. Let $\tilde{\Delta}_n^i := c_{n,r}^i - c_{n,v}^i$, $\tilde{x}_1(Q) := (L_n^{i'})^{-1}(\tilde{\Delta}_n^i) - \min\{Q, 0\}$ and $\tilde{x}_2(Q) := (L_n^{i'})^{-1}(\tilde{\Delta}_n^i) - \max\{Q - C_{n,r}^i, -C_{n,v}^i\} > \tilde{x}_1(Q)$.*

(a) If $x \leq \tilde{x}_1$:

$$q_r^* = \max\{Q, 0\}, \quad q_v^* = \min\{Q, 0\}.$$

(b) If $\tilde{x}_1 < x < \tilde{x}_2$:

$$q_r^* = Q + x - (L_n^{i'})^{-1}(\tilde{\Delta}_n^i), \quad q_v^* = (L_n^{i'})^{-1}(\tilde{\Delta}_n^i) - x.$$

(c) If $x \geq \tilde{x}_2$:

$$q_r^* = \min\{Q + C_{n,v}^i, C_{n,r}^i\}, \quad q_v^* = \max\{Q - C_{n,r}^i, -C_{n,v}^i\}.$$

(d) $G_{n,2}^i(x, Q)$ is continuously differentiable with respect to x , and

- for $x \leq \tilde{x}_1(Q)$, $\frac{\partial G_{n,2}^i(x,Q)}{\partial x} = L_n^{i'}(x + \min\{Q, 0\})$;
- for $\tilde{x}_1(Q) < x < \tilde{x}_2(Q)$, $\frac{\partial G_{n,2}^i(x,Q)}{\partial x} = \tilde{\Delta}_n^i$;
- for $x \geq \tilde{x}_2(Q)$, $\frac{\partial G_{n,2}^i(x,Q)}{\partial x} = L_n^{i'}(x + \max\{Q - C_{n,r}^i, -C_{n,v}^i\})$.

C Sequential Ordering Where the Regular Supplier Goes First

Sometimes, under sequential ordering, one order can be placed with the regular supplier first, where realized delivery becomes revealed prior to having to decide on any supplementary expedited order. This setting can be

accommodated like the one in the base model, with a few minor alterations to the assumptions.

In order to maintain one-dimensional dynamic programs, and analogous structural results as in Section 4, we now assume that the lead time for salvaging is $l_v = l_r$ (if salvaging applies). The realization of the random supply mechanism $\hat{Y}_{n,r}(\mu_r)$ in period n becomes revealed at the end of the period of placement of that order, after which we may decide on a supplementary expedited order to arrive at the beginning of period $n - l_v = n - l_r$, together with the regular order (and the salvage quantity). The new formulation of the sequential dynamic program then follows:

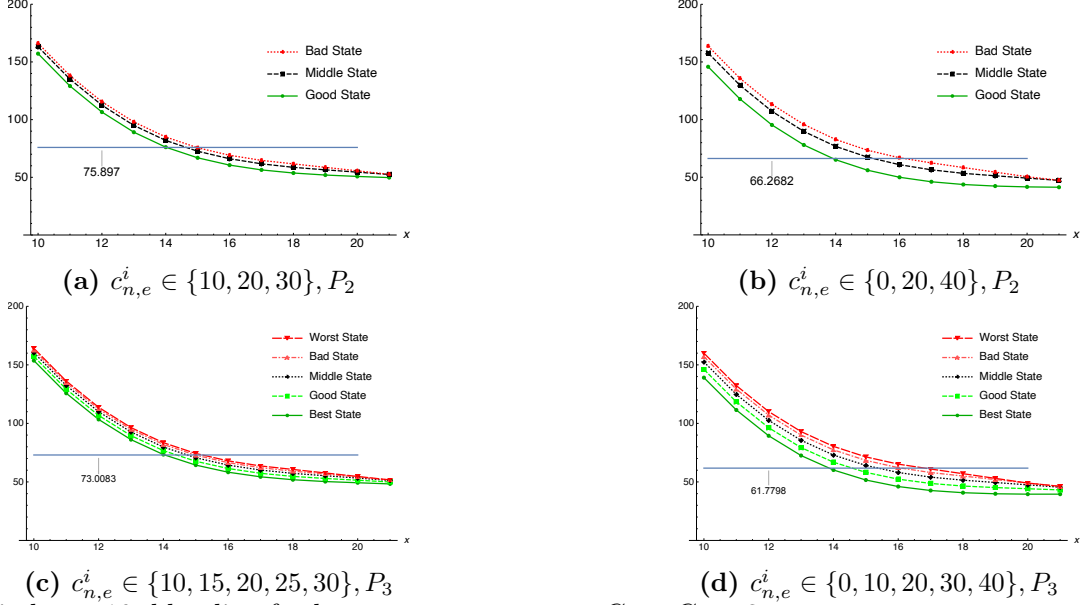
$$f_n^i(x) = \min_{-C_{n,v}^i \leq q_v \leq 0, \mu_r \in \mathcal{U}_{n,r}^i} \left\{ c_{n,v}^i q_v + c_{n,r}^i \mu_r + \mathbb{E} f_{n,e}^i(x + q_v + \hat{Y}_{n,r}^i(\mu_r)) \right\},$$

where

$$f_{n,e}^i(y) = \min_{\mu_e \in \mathcal{U}_{n,e}^i} \left\{ c_{n,e}^i \mu_e + \mathbb{E} L_n^i(y + \hat{Y}_{n,e}^i(\mu_e)) + \alpha \sum_{j=1}^I p_{ij} \mathbb{E} f_{n-1}^j(y + \hat{Y}_{n,e}^i(\mu_e) - D_n^i) \right\}.$$

D Additional Numerical Results

This appendix supplements the numerical studies in Section 6.3. First, Figure 4 displays the optimal cost-to-go functions, with a remaining planning horizon of $n = 10$. We choose the two Markov chains P_2 and P_3 from the first part of Table 2.



Note. Period $n = 10$; blue line for long-run average cost; $C_e = C_r = 3$.

Figure 4: Optimal cost-to-go under different states-of-the-world and inventory positions

To conduct the systematic study comparing the long-run average costs, as a function of β and the total variation distance (see Section 6.3), we use the following parameterized transition matrix and cost rate vector for the expedited supplier.

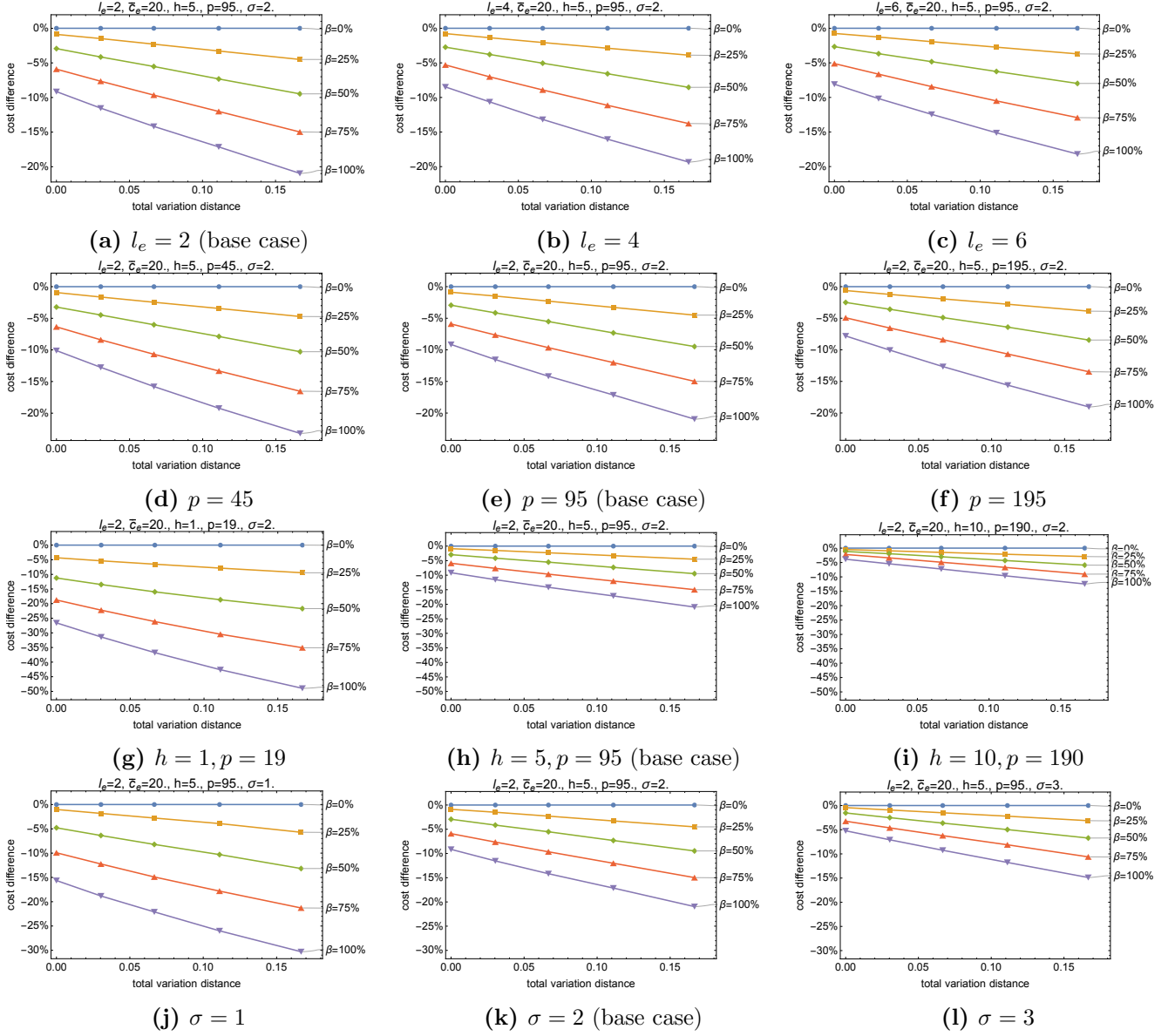
$$P(a) = \begin{pmatrix} 1/3 + a & 1/3 & 1/3 - a \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} \Rightarrow \pi(a) = \left\{ \frac{1}{3-3a}, \frac{1}{3}, \frac{1-2a}{3-3a} \right\} \quad (46)$$

$$\Rightarrow \text{total variation distance}(a) = \frac{1}{2} \left\| \pi(a) - \left\{ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\} \right\|_1 = \frac{a}{3-3a}$$

with $0 \leq a \leq \frac{1}{3}$, so that the total variation distance increases from 0 to $\frac{1}{6}$ in a . As defined in Section 6.3,

$$c_{n,e}(\beta) = \{20(1 - \beta), 20, 20(1 + \beta)\}.$$

In each subfigure of Figure 5, we put the total variation distance on the horizontal axis, displaying the cost savings vis-a-vis the base case ($\beta = 0$) on the vertical axis. We exhibit 5 curves for increasing values of β .



Note. Common parameters: $C_e = C_r = 3$, $c_{n,r}^i = 0$, $\mu = 5$, $\alpha = 1$.

Figure 5: Optimal long-run average cost reduction from varying Markov chain and unit expedited cost

We now investigate whether and how the volatilities in the capacity level can be exploited to achieve additional cost savings. We fix $C_e = 3$, $c_{n,e}^i = 20$ and $c_{n,r}^i = 0$ while varying the regular supplier's capacity limits. Table 3 compares the optimal long-run average cost for systems with different Markov chains and different Markov-modulated capacity limits for the *regular* supplier. We consider the two Markov chains P_1 and P_2 below. Under each Markov system, three sets of capacity limits for the regular supplier are evaluated, all centered around $C_r = 3$, the capacity level in the “middle state”.

Table 3: Optimal long-run average cost for different Markov-modulated systems

Transition matrix C_r	P_1		P_2	
	cost	diff ¹	cost	diff ¹
{3, 3, 3}	83.84		83.84	
{4, 3, 2}	86.38	+3.0%	75.94	-9.4%
{5, 3, 1}	96.59	+15.2%	75.10	-10.4%

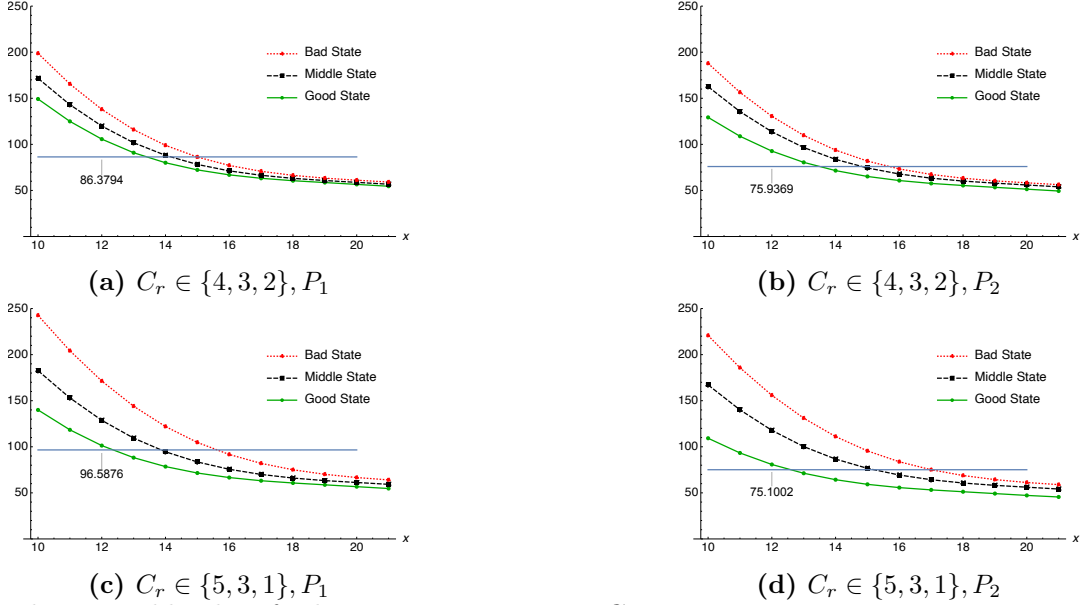
$$P_1 = \{\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}, \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}, \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}\}$$

$$P_2 = \{\{\frac{2}{3}, \frac{1}{3}, 0\}, \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}, \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}\}$$

$$C_e = 3, c_{n,e}^i = 20, c_{n,r}^i = 0$$

¹ Relative difference to the (bold) benchmark costs

Comparing rows in Table 3 under each Markov chain, we find that the optimal long-run average cost is increasing when C_r fluctuates more under P_1 but the *opposite* occurs under P_2 . This indicates that the impact of increased volatility in the regular supplier's capacity depends on the Markov chain: under the equal-transition probability matrix P_1 , the cost increases with the volatility of C_r , but it decreases with the volatility of C_r under P_2 , where the likelihood of staying in the good state (with high capacity) is 2/3, as opposed to 1/3.



Note. Period $n = 10$; blue line for long-run average cost; $C_e = 3, c_e = 20, c_r = 0$.

Figure 6: Optimal cost-to-go under different states-of-the-world and inventory positions

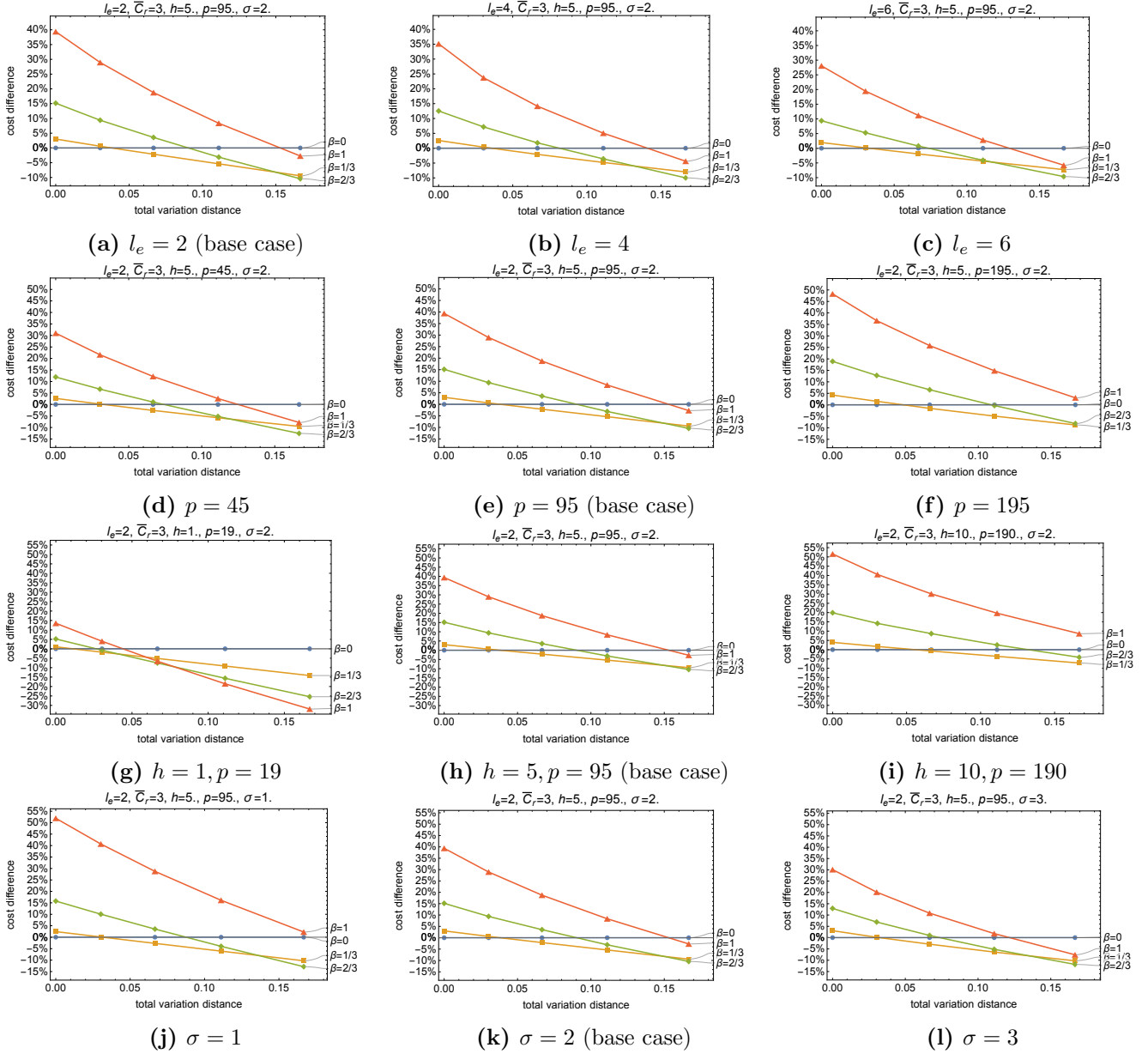
Figure 6 shows the value function $f_n^i(x)$ in four instances, selected from those in Table 3, all relating to period $n = 10$. In each subfigure, it is clear that the cost-to-go is ranked by the state-of-the-world for any inventory position x , illustrating Theorem 5. The comparisons of subfigures (a) vs (b) and (c) vs (d) both show the value function is lower under P_2 ((b) and (d)) than under P_1 ((a) and (c)), since the transition probability of staying in the good state is higher under P_2 , thus illustrating Proposition 4.

As in Section 6.3, we systematically investigate the cost savings (positive or negative), as a function of the “total variation distance” and $\frac{\frac{1}{2}\text{span}\{C_r^1, \dots, C_r^I\}}{\text{avg}\{C_r^1, \dots, C_r^I\}}$, the analogue of the dispersion measure in Section 6.3. To this end, we employ the same family of Markov chains as in (46), and the parameterized capacity vector

$$C_r(\beta) = \{3(1 + \beta), 3, 3(1 - \beta)\}.$$

Figure 7 is the complete analogue of Figure 4. It confirms the above observations following Table 3. A higher

“total variation distance” implies a larger steady state probability of being in the good state, therefore a better cost performance. As to the dependency on β , the cost implications are no longer uniform, which is why the curves within the same graph are sometimes crossing, and the cost “savings” sometimes positive and sometimes negative.



Note. Common parameters: $C_e = 3, c_{n,e}^i = 20, c_{n,r}^i = 0, \mu = 5, \alpha = 1$.

Figure 7: Optimal long-run average cost reduction from varying Markov chain and regular capacity

Online Appendix

OA.1 Problems with Feng and Shanthikumar (2018)

In this appendix, we detail the issues we identify with Feng and Shanthikumar (2018).

Feng and Shanthikumar (2018) aims to provide a unified treatment for inventory problems with general random supply and demand functions. They introduce the notion of *stochastic linearity in midpoint* (SL(mp)), which is the key to ensure that the DP value functions maintain concavity properties. Theorem 4 of Feng and Shanthikumar (2018) aims to establish the optimality of a threshold policy for inventory problems satisfying the SL(mp) property. However, we find various places in its proof (in the online appendix of Feng and Shanthikumar (2018)) that are either flawed or wrong. We detail them below.

1. The inequality at the top of page 40 (copied here),

$$\begin{aligned} & \frac{\mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3) + \Delta), x_3)] - \mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3)), x_3)]}{\Delta} \\ & \leq \frac{\mathbb{E}[\phi(x_3 + \Delta + S_j(\mu_{S_j}^*(x_3)), x_3)] - \mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3)), x_3)]}{\Delta} \leq c_j, \end{aligned}$$

is, clearly, equivalent to (37) (copied here)

$$\mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3) + \Delta), x_3)] - c_j(\mu_{S_j}^*(x_3) + \Delta) < \mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3)), x_3)] - c_j\mu_{S_j}^*(x_3),$$

as follows simply by rearranging the terms in (37) and dividing both sides by Δ . It therefore seems to confirm (37) rather than “contradicts (37)” as stated in the paper. Therefore (38) cannot be rejected as a hypothesis, and its “opposite” (39) is not established.

2. The inequality string, at the top of page. 40, employs a first inequality (copied here)

$$\begin{aligned} & \frac{\mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3) + \Delta), x_3)] - \mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3)), x_3)]}{\Delta} \\ & \leq \frac{\mathbb{E}[\phi(x_3 + \Delta + S_j(\mu_{S_j}^*(x_3)), x_3)] - \mathbb{E}[\phi(x_3 + S_j(\mu_{S_j}^*(x_3)), x_3)]}{\Delta} \end{aligned}$$

which is justified by Lemma 6 (copied here);

Lemma 6. *If $\{Y(x), x \in \mathcal{X} = [x, \bar{x}] \subseteq [0, \infty)\} \in \text{SI}(\text{disp})$ and $\mathbb{E}[Y(x)] = x$, then*

$$\lim_{\Delta \downarrow 0} \frac{\mathbb{E}[\phi(Y(x + \Delta))] - \mathbb{E}[\phi(Y(x))]}{\Delta} \leq \left. \frac{\partial \mathbb{E}[\phi(Y(x) + a)]}{\partial a} \right|_{a \downarrow 0}$$

for any concave $\phi(\cdot)$.

However, Lemma 6 ranks the limits for $\Delta \downarrow 0$ rather than a global ranking for general Δ , sufficiently small. Therefore, this first inequality in this string remains unsubstantiated.

3. In the statement of Lemma 6, in (34) (copied below)

$$\lim_{\Delta \downarrow 0} \frac{\mathbb{E}[\phi(Y(x + \Delta))] - \mathbb{E}[\phi(Y(x))]}{\Delta} \leq \frac{\partial \mathbb{E}[\phi(Y(x) + a)]}{\partial a} \Big|_{a \downarrow 0},$$

it is not clear if the right-hand-side is intended to represent the partial derivative at $a = 0$ or the limit of this partial derivative as $a \downarrow 0$. The notation suggests the latter; indeed, for general concave functions, the partial derivative may fail to exist in specific points. But if so, there is a second limit process, letting $a \downarrow 0$, in addition to that pertaining to $\Delta \downarrow 0$, further complicating its representation with general finite differences in the first inequality of the above-mentioned string.

4. In the middle of page 40, the envelope theorem is invoked to evaluate the derivative of the V function in $\hat{x}$ (see below)

$$\begin{aligned} \frac{dV_t(x)}{dx} \Big|_{x=\hat{x}} &= \frac{\partial \mathbb{E}[\phi(x + S_j(\mu_{S_j}^*(\hat{x})), \hat{x})]}{\partial x} \Big|_{x=\hat{x}} \\ &= \frac{d\phi(x, \hat{x})}{dx} \Big|_{x=\hat{x}} = \lim_{\Delta \downarrow 0} \frac{\mathbb{E}[\phi(\hat{x} + S_j(\Delta), \hat{x})] - \mathbb{E}[\phi(\hat{x} + S_j(0), \hat{x})]}{\Delta} \leq c_j. \end{aligned}$$

However, the expression omits the partial derivative of $\mathbb{E}H(x - D)$ with respect to x , see (13)–(14) (copied below)

$$\begin{aligned} V_t(x_t) &= \max_{\mu_D \leq \mu_D \leq \bar{\mu}_D, 0 \leq \mu_{S_i} \leq \bar{\mu}_{S_i}, i \in \mathcal{N}} \hat{J}_t(x_t, \mu_D, \mu_{S_1}, \dots, \mu_{S_n}), \\ \hat{J}_t(x_t, \mu_D, \mu_{S_1}, \dots, \mu_{S_n}) &= p(\mu_D)\mu_D - \sum_{i \in \mathcal{N}} c_i \mu_{S_i} - \mathbb{E}[H(x_t - D(\mu_D))] + \rho \mathbb{E} \left[V_{t+1} \left(x_t - D(\mu_D) + \sum_{i \in \mathcal{N}} S_i(\mu_{S_i}) \right) \right]. \end{aligned}$$

5. Continuing with the above derivative expression, in the finite difference expression in Δ , a Δ -term needs to be added to the argument of the first $\phi(\cdot)$ term.
6. Finally, on page 37 the authors announce that the proof of Theorem 4 is based on Lemma 7 (as well as lemma 6). However, it cannot be found at which place in the proof of Theorem 4 this Lemma 7 is invoked.

OA.2 On Tan et al. (2016)

Proposition 1 in Tan et al. (2016) states (a) the optimality of individual threshold policies along with (b) various inequalities for the optimal order sizes. However, the proof does not relate at all to (a) and seems to take the structure for granted. Moreover the proof starts with equations (2) and (3) *ibid.*, which do not apply in general, but only for interior solutions. The essence of the structure of an optimal policy is, in fact, which optimal order sizes are interior to the feasible region or at their boundary and equal to zero.